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IN THE WORLD OF SCIENCE

Практикум

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Учебный материал включает тексты, затрагивающие такие области науки, как физика, математика, химия, биология. Задания направлены на контроль понимания прочитанного, усвоение новой лексики, совершенствование навыков учебного перевода.

Предназначен для проведения практических занятий на факультетах и институтах, осуществляющих подготовку будущих специалистов технических специальностей.

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UNIT 1. SCIENCE IN OUR LIFE

Lead-in:

1. What is the difference between science and technology?
2. Who are the most famous scientists from our country? What did they do?
3. How will science change the world in the next 100 years?
4. In your opinion, what is the greatest technological invention? Why?
5. How have technological advances changed our life? Are they always good?

What is science?

1. A) Read and translate the text.

Science consists of studying the world by watching, listening, *observing,* and *recording*. Anyone can have an idea about how nature works. Some people think their *idea* is correct because «it seems right» or «it makes sense» But for a *scientist*, this is not enough. A scientist will test the idea in the real world.

An idea that predicts how the world works is called a *hypothesis*. If an idea, or hypothesis, correctly predicts how something will behave, we call it a *theory*. If an idea explains all the facts, or *evidence*, that we have found, we also call it a theory.

Scientific *method* usually means a series of particular steps that scientists follow to discover how nature works. But this is not usually the way science actually happens! Sometimes the *observations* come before the idea or theory. Sometimes science happens mostly inside a scientist's head. Albert Einstein and his theories were like that. It took a long time before scientists were able to test them and show that they were correct.

B) Match the words in italics with their correct definitions.

1. watching carefully the way something happens or the way someone does something, especially in order to learn more about it;
2. anything that helps to prove that something is or is not true;
3. knowledge from the careful study of the structure and behaviour of the physical world;
4. an idea or explanation for something that may be true based on known facts but has not been proved;
5. keeping information for the future, by writing it down or storing it on a computer;
6. a remark made about something you have noticed;

7. a particular way of doing something;

8. a statement of ideas that explain a fact or event, or how something works;

9. an understanding, thought, or picture in your mind;

10. an expert who studies or works in one of the fields of science.

C) Find in the text the English equivalents for the following Russian words and phrases.

Изучать окружающий мир; иметь смысл; проверять идеи; предсказывать, или предполагать; объяснить все факты; ряд определенных шагов; обнаружить, или узнать; подтвердить правильность теории.

2. A) Read the text about the fields of science.

The *field of «science»* is often grouped into:

• *natural science* — life or biological science (the study of *living organisms*) and physical science (the study of the *material universe* including physics, chemistry, space science etc);

• social science — the study of society and people (such as anthropology, psychology);

• formal science — the study of logic and mathematics;

• *applied science* — disciplines that use existing *scientific knowledge* to develop new *applications*, such as in engineering, robotics, agriculture and medicine.

Both natural science and social science are known as *empirical* sciences. This means that any theories must be based on *scientific research*, *observable phenomena*, *reproducibility of results* and *peer review*.

B) Find in the text the English equivalents of the following:

Наблюдаемые явления; естествознание; прикладные науки; основанный на опыте, полученный опытным путем (эмпирический); область науки; материальная вселенная; научные знания; научное исследование; воспроизводимость результатов; живые организмы; практическое применение; экспертная оценка коллег.

3. A) Sort out the following fields of science by putting them into four categories.

psychology, ecology, chemistry, economics, robotics, genetics, physics, meteorology, neuroscience, microbiology, engineering, mathematics, history, computer science, law, statistics, geology, space science, logic, pedagogy, medicine, astronomy

NATURAL SCIENCE	SOCIAL SCIENCE	FORMAL SCIENCE	APPLIED SCIENCE

B) What does each discipline study? Answer the question using the given model.

Model: Geography is a natural science. It studies lands and different Earth's phenomena.

C) Form adjectives and nouns from the words in part A using the given model.

Model: psychology — psychological — psychologist (subject — adjective — person)

4. A) Decide whether the information about seven principles of science are true or false. Correct mistakes.

Principle 1: *Universal open access*

Everyone should have open access to *scientific records* to read and copy without any changes.

Principle 2: *Open licensing*

Scientific works should have open licenses that give the writer *copyrights*.

Principle 3. *Peer review*

It is important to have careful and constant peer review as a part of any *research*.

Principle 4. *Supporting metadata*

General public and scientists cannot see *all the data and evidence*.

Principle 5. Access by future generations

Important data and information should be a secret for *future generations*.

Principle 6. Respecting traditions

There are specific traditions of different disciplines which scientists need to follow.

Principle 7. *Creating opportunities*

Scientists should not give other people any chances to *overcome new challenges* and *solve new scientific questions*.

B) Explain the meaning of the phrases in italics. Translate the phrases.

5. A) Read a short text about the difference between scientists and inventors.

What is the difference between a *scientist* and an *inventor*? In general terms a scientist is a person who *studies nature* and *makes theories and discoveries* as to how nature works *using scientific methods*. An inventor takes *the laws and theories of science* and *puts them to practical use* by humans. Many people are both scientists and inventors. For example, Isaac Newton was a scientist when he wrote about the theory

of gravity, but he was also an inventor when he made the first working reflecting telescope.

B) Find in the text the English equivalents of the following:

Находить практическое применение; строить теории; ученый; изучать природу; научные законы; совершать открытия; изобретатель; применять научные методы; научные теории.

C) Project work: «Famous scientists and inventors».

Give a talk about a famous scientist and a famous inventor. Do not forget to mention their contribution to the history of science.

Basic scientific terms

6. A) Translate basic scientific terms on the left. Match them with their translations on the right.

assessment	a) a number, *amount*, or situation that can change;
accuracy	b) a *particular* number or amount that never changes;
chart	c) *a set of numbers*, amounts, etc., used *to measure or compare* the level of something;
compound	d) the fact of being *correct* and without any mistakes;
configuration	e) a drawing that shows information using *lines and curves* to show amounts;
constant	f) a very small *piece of matter* that together with others *makes an atom*;
data	g) the amount or number of something, or a particular amount or number;
formula	e) how *useful* or important something is;
matter	f) the particular *arrangement of the parts* of something or of a group of things;
particle	g) something *consisting of* two or more different parts;
quality	h) the *relationship* between two groups or amounts that *expresses* how much bigger one is than the other;
quantity	i) *physical substanc*e in the *universe*;
ratio	j) a word or *expression used to describe* something official or technical;
scale	k) the *process of testing*, and *making a judgment* about something or someone;
substance	l) information, especially facts or numbers, collected *for use*;
surface	m) a *standard* of how good or bad something is;
term	n) a *formal statement* that scientists *prove*;
theorem	o) a standard *way of doing or making something*;
value	p) a material with particular *physical characteristics*;
variable	q) a top part or *layer* of something.

B) Translate the words and phrases in italics into Russian.

C) Make up your own sentences using basic scientific terms.

7. Choose the correct answer. Translate the sentences.

quality – quantity – value

The main difference between quality and quantity is the fact that ____ refers to the characteristic or feature of something, whereas ____ refers to the numerical value of something. When you think about the usefulness and practicality of something, you consider its ____.

term – theorem – formula

____ is a rule or law which scientists need to prove. ____ is a word or expression with a particular meaning in some fields of science, art or profession. ____ is a fact or a rule written with mathematical symbols.

particle – matter – substance

All ____ is made of tiny ____. ____ is a matter with particular composition and features.

ratio – scale

____ is a set of numbers or amounts used to measure or compare the level or something. ____ is the relationship in quantity, amount or size between two or more things.

variable – constant

A ____ does not change its value and it remains the same forever. A ____, on the other hand, changes its value from time to time.

accuracy – assessment

The synonym to ____ is correctness, precision. ____ is a process of systematic collection, review or use of information.

data – chart

If you want to present ____ using diagrams or graphs, it is better to make a ____. Individual facts, statistics or some pieces of information are called ____.

configuration – compound – surface

Top part of something is called a ____. ____ is an arrangement of parts or elements. When a substance is made from two or more elements, it is a ____.

8. A) Match the words and phrases with their Russian equivalents.

do experiments; *do research*; *solve problems*; *disprove a theory; hypothesize*; *invent*; *collect data*; *make theories*; *make predictions*; *undermine a theory; sustain a theory*

Находить решение проблемы; опровергнуть теорию; ставить под сомнения теорию; собирать данные; проводить эксперименты; создавать, или строить теории; подтверждать теорию; выдвигать гипотезу; прогнозировать; изобретать; проводить исследование.

B) Make up your own sentences using the vocabulary from part A.

9. A) Fill in the blanks with the proper words from the box.

test	correct	discover	evolution	repeat	support	experiments
incorrect	laboratory	works	theory	telescope	undermine	journal

The aim of science is to ____ new facts about the world. Scientists *find out new facts* by *making observations*, or by doing ____. Astronomers, for example, discover new facts by making observations through a ____, and chemists discover new facts by doing experiments in a ____.

An idea that explains why something is true, or how something happens, is called a scientific ____. For example, Charles Darwin *created the theory* of ____ to explain how all the different species in nature developed over time. Scientists then *designed experiments* and made observations in order to ____ his theory.

The results of an experiment might sustain a theory or they might ____ it. If scientists keep getting results that *weaken a theory*, they should *accept* that it is ____. But if they *keep getting results* that ____ a theory, they should accept that the theory is ____.

Scientists who find new *evidence that supports or undermines a theory* will *publish* their *results* in a scientific ____. The results will only be *accepted as valid* if others can ____ the observation or experiment and *get the same results*. This is how science ____.

B) Explain the meaning of the words and phrases in italics. Translate them into Russian.

Scientific breakthroughs

10. A) Skim the text about 10 interesting scientific discoveries. Match headings to the paragraphs.

a) Baby mammoth discovered in Canadian ice; b) Ice Age-era footprints; c) Knocking asteroids; d) Brain cells playing video games; e) People in coma hearing us; f) Greenland is quickly disappearing; g) Near-Earth asteroids; h) One billion crabs lost; i) Baby rats with brain cells; j) Men age faster than women

B) Read and translate the text paying attention to the phrases in italics. When can we use them?

The 10 Most «OMG» Science Discoveries of 2022

1. *Researchers* in Finland *looked at* 2,240 twins in two age groups: those between 21 and 42 years of age and those between 50 and 76. Using the epigenetic clock, a biochemical test used to measure age, *the scientists compared* each person's chronological age to how old the epigenetic clock said they are biologically. *The study author said that* when

comparing male-female twins, the male was about one year biologically older than his sister in his 20s and four years in his 50s.

2. NASA smashed a spacecraft, known as DART, directly into an asteroid in September. *Their aim was to see* if such a collision could knock the asteroid off its orbit, something that could protect Earth from an apocalyptic asteroid strike, like the one that wiped out the dinosaurs millions of years ago. It slammed into the space rock at 14,000 miles an hour and was destroyed instantly.

3. «Covert consciousness» is a state in which the brain reacts to the outside world with some understanding, but the body remains inactive. *It is reported that* as many as 15 to 20 percent of patients in a coma have this kind of internal understanding when monitored with technology that can measure brain activity.

4. The world's second-largest sheet of ice, also known as Greenland, *seems to be* disappearing faster than scientists previously thought. Warmer ocean waters and rising air temperatures have accelerated the Arctic land's melting. *According to a study published in the journal* Nature Geoscience, Greenland is losing about 250 billion metric tons of ice each year.

5. When miners in Canada discovered an animal frozen within the permafrost, they quickly called experts. None of them were prepared for what happened: University of Calgary researchers were shocked when *it was determined to be* a female baby mammoth, about 30,000 years old, with perfectly preserved toenails, skin, trunk, and hair — the best-preserved mammoth ever found in North America.

6. *In a study published in the NPR journal* this October, scientists injected human nerve cells into the brains of rats. *They found that* those neurons continued to grow, forming connections with their host's brain cells and guiding their behavior. «*The goal of this work is* to begin to understand features of complex diseases» Paola Arlotta, a Harvard neuroscientist, told NPR.

7. Australian researchers say they've grown brain cells in a lab that have learned to play the vintage video game Pong. The «mini-brains» they created can sense and respond to their environment. Dr. Brett Kagan said his team had created the first «feeling something» brain grown in a lab. The mini-brain learned to play the game in five minutes, the researchers said. It missed the ball sometimes, but its understanding of the game was better than the random chance.

8. *A study* published in August *revealed that* scientists had found 88 footprints belonging to adults and children, likely dating back 12,000 years, in Utah. *They suggest* humans occupied the area 7,500 years ear-

lier *than previously thought*. «Now that we have this human element, the story of the very early people becomes more real» David Madsen, an archaeologist at the University of Nevada-Reno, told CNN. It is a real scientific breakthrough.

9. The European Space Agency says there are more than 30,000 near-earth asteroids (NEAs) in the solar system. They're space rocks that revolve around the sun on paths relatively close to Earth's orbit. And 1,425 of them have a «non-zero chance» of hitting the earth and *they are being watched closely* by astronomers. On average, Earth is hit by a large asteroid every 5,000 years and a civilization-ending asteroid every one million years, says NASA.

10. In October, CBS News *reported that* one billion crabs have disappeared from Alaska in the last two years, and experts aren't sure why. That amounts to 90% of their population. Disease is a potential explanation. Climate change is another. Alaska is the fastest-warming state in the U.S, and crabs need cold water to survive. Miranda Westphal, a biologist with Alaska's fish department, said that between 2018 and 2019, the Bering Sea «was extremely warm and the snow crab population moved to the coolest water they could find».

B) Answer the following questions.

1. Why is it said in the article that men age faster than women? 2. Why did Nasa smash a spacecraft into an asteroid? 3. What is «covert consciousness»? 4. What can happen if Greenland continues losing 250 billion metric tons of ice? 5. Why was the mammoth found in Canada the best-preserved mammoth ever? 6. What happened when human nerve cells were injected into the brains of rats? 7. How much time did grown brain cells need to learn how to play a computer game? 8. Why did David Madsen say that the story of the very early people became real? 9. What is a near-earth asteroid? 10. What are the reasons for one billion crabs disappearing?

C) Find in the text the English equivalents of the following:

Научное открытие; определить возраст; при сравнении; сбить астероид с орбиты; падение, или удар астероида; мгновенно уничтожить; состояние; отслеживать при помощи оборудования; активность мозга; ученые ранее думали; ускорить таяние ледников; позвать на помощь экспертов; идеально сохранившиеся; формировать связи; понимать особенности сложных заболеваний; выращенный в лаборатории; случайность, или произвольные изменения; датируемый; прорыв в науке; солнечная система; вращаться вокруг солнца; быть эквивалентным; выжить.

D) Retell the text.

11. Project work: «The most interesting scientific discoveries and inventions».

In pairs make a presentation about 5 interesting scientific discoveries in different fields of science. Use phrases in italics from exercise 10 to talk about them.

REVISION

1. Circle the correct word. Translate the sentences into Russian.

1. Without the *discovery* / *invention* / *making* of penicillin, we would be living in a very different world right now.
2. What do we call a *scientist* / *science* / *scientific* who studies physics?
3. Scientists carry out *research* / *experiments* / *discoveries* in laboratories.
4. *Substance* / *state* / *scale* is a material with particular physical characteristics.
5. Vaccines were a major *fact* / *breakthrough* / *science* in the fight against diseases.
6. Isaac Newton's experiments *pointed* / *undermined* / *proved* his theory that gravity existed.
7. A *variant* / *various* / *variable* is a number, amount, or situation that can change, while a *consonant* / *conscious* / *constant* is a particular number or amount that never changes.
8. An idea that is based on a few known facts, but that hasn't been proved yet, is called *analysis* / *hypothesis* / *theorem*.
9. Scientists *scan* / *view* / *observe* an object or an activity.
10. Fire was the *first invention* / *discovery* / *findings* and wheel was the first *invention* / *discovery* / *findings*.

2. Say what you can remember about:

1. scientific ideas and their differences;
2. scientific methods and ways to study the world;
3. main science fields and the difference between them;
4. principles of science;
5. the differences between a scientist and an inventor;
6. the way science works;
7. some major breakthroughs in recent years.

3. Translate from Russian into English.

1. *Ученые* изучают окружающий мир, применяя *научный метод*: в ходе *исследования* ученый *проверяет идеи*, *объясняет факты*, *строит теории* и *ставит* их *под сомнение*.

2. *Представить* все данные, *наблюдения* и *доказательства* лучше *в виде графика*.

3. Чтобы *подтвердить теорию*, необходимо *выдвинуть гипотезу*, затем *собрать данные и доказательства*, *провести эксперимент*.

4. Любой ученый должен соблюдать основные *научные принципы*: предоставлять *общий открытый доступ* к научным *записям*, отдавать свое *исследование* на *экспертную оценку коллег*, соблюдать правила работы в конкретной *научной области*, не нарушать *авторские права* других *исследователей* и давать возможность другим ученым *решать* новые *научные задачи*.

5. *Изобретатель находит практическое применение* существующим *научным законам* и теориям.

6. Все *переменные величины* принято обозначать буквами греческого алфавита. *Величины постоянные* обозначают латинскими буквами.

7. *Результаты исследования* могут *считаться действительными* только в том случае, если другие ученые смогут *повторить данный эксперимент* и получить те же данные. Таким образом, *воспроизводимость результатов* – основа для *подтверждения правильности теории*.

8. *По данным исследования, опубликованного в журнале* «Science», столкновение крупного астероида с Землей происходит каждые 5000 лет.

9. *Целью данного исследования* было проследить, как нейроны влияют на активность мозга в процессе обучения. Работа нейронной сети мозга связана не только с восприятием и пониманием информации, *как считалось ранее*. *Сообщается*, что от нейронов также зависит то, как мы читаем буквы и складываем их в слова.

10. В 2022 году международная группа ученых при участии проекта «Радиоастрон» *совершила* невероятное *научное открытие*: была *обнаружена* система OJ 287 – на сегодняшний день единственная тесная двойная черная дыра (narrow double black hole). Это стало настоящим *прорывом* в *астрономии*.

UNIT 2. PHYSICS

Lead-in:

1. How old is the science of physics? What does it study?
2. Can you name any famous physicists and their important discoveries?
3. What are the main branches of physics?
4. What breakthroughs laid the foundations of the 'new era' in physics?
5. What is the link between physics, technology and society?

Text 1. A brief history of physics

1. Read the text and choose the best title for each paragraph. There is one title which you do not need to use:

A. Science in Classical antiquity B. Deep into the Atom C. Challenges nowadays D. The ancient period E. Ideas that speeded up the industrial revolution F. Scientific revolution and the origin of modern physics

Although the history of physics as an independent science began only in the 17th century, its *origins* date back to the deepest antiquity. The accumulation of astronomical knowledge could not but affect the development of mathematics. **1**_____They attributed lightning and solar *eclipses* to divine wrath, which had nothing to do with science.

The word physics itself is derived from the Greek word *phusis* meaning nature. In the classical period in Greece, a genius philosopher Aristotle revealed the importance of observation and considered it the key to discovering the laws that control natural phenomena. **2**_____He wrote the first work «Physics» in the 4th century BC and formed the theory of five elements. The great mathematician and inventor from Alexandria Archimedes developed the principles of *equilibrium* states and the centers of *gravity*. His best-known achievement was «Eureka» moment, when he discovered the principle of *buoyancy*.

Eastern *scholars* from India, Egypt and China were able to use calculations to predict the movement of the Sun or build machines. There were studies of planetary motion by Indian astronomers, the theories of light from Buddhist and Persian thinkers and especially the work of the Persian philosopher Nasir al-Din al-Tusi on the planetary system. Their work reached Europe in the 12th and 13th centuries and eventually, these ideas pushed Europe into a scientific revolution. Instead of accepted then geocentric Nicolaus Copernicus introduced the *heliocentric* system claiming that the planets move around the sun. Later

Johannes Kepler concluded that the planets follow not *circular* but elliptical orbits and formulated three laws of planetary motion. **3**_____His astronomical discoveries and his work in mechanics *foreshadowed* the work of the 17th century English mathematician and physicist Sir Isaac Newton who set out the Laws of Motion, devised the basic ideas of *calculus* and modern physics was born.

The next great area of *investigation* was electricity. In the 19th century Michael Faraday discovered electromagnetic induction and James Maxwell stated Faraday's physical ideas into mathematical form, showing that light was electromagnetic radiation. In proving Maxwell's *equations*, Heinrich Hertz discovered radio waves and Wilhelm von Roentgen, X-rays. Maxwell's work was also the starting point for Einstein's Theory of Relativity. **4**_____Physicists such as Robert Boyle, James Prescott Joule and many others set out the theories that allow us today to make use of engines and other mechanical devices. Roentgen's discovery of X-rays and the work of Pierre and Marie Curie on radioactivity led to the development of the science of nuclear physics.

In the first half of the 20th century, developments in physics were concerned with the structure of atoms. **5**_____Also at that time, German scientist Max Planck proposed that energy is radiated in very *minute* and discrete quantized amounts or packets, rather than in a continuous wave. *Quanta* were the modern answer to Newtonian theories. The field of quantum mechanics, which explains not only how atomic particles move, but how the universe does, came into being. Without physics to describe the way things behave, we would have none of the technology and machinery we take for granted today.

2. The following sentences are cut from the text. Decide where they fit:

1. At the same time, other scientists were working on thermodynamics, that is, the study of changes of heat in matter. 2. Galileo Galilei constructed his own telescope in 1609 and by observing the phases of the planet Venus confirmed the heliocentric system. 3. However, the ancients could not explain the causes of natural phenomena theoretically. 4. The parts of the atom were identified – its *nucleus*, protons and electrons. 5. He viewed the process of learning as one of observation and thinking, but he would not conduct experimentation.

3. Match the following names of the prominent scientists and their achievements:

1. Albert Einstein	a. Astronomical tables for planetary movements
2. Archimedes	b. Three fundamental laws of planetary motion

3. Aristotle	c. The concept of a physical experiment; Law of inertia; the phases of Venus
4. Galileo Galilei	d. Studies on the nature of heat; The theory of conservation of energy and the first law of thermodynamics
5. Heinrich Hertz	e. X-rays
6. Isaac Newton	f. Laws of electromagnetic induction
7. James Maxwell	g. Principle of buoyancy; Principle of the levers
8. James Joule	h. Studies on natural radioactivity; Discovery of polonium and radium
9. Johannes Kepler	i. Heliocentric view of the solar system
10. Max Planck	j. Explanation of photoelectric effect; Theory of relativity
11. Michael Faraday	k. Universal law of gravitation; Laws of motion; Reflecting telescope
12. Nasir al-Din al-Tusi	l. The concept of energy quanta
13.Nicolaus Copernicus	m. Generation of electromagnetic waves
14. Marie Curie	n. The theory of the elements
15. W.K. Roentgen	o. Electromagnetic waves; Light – an electromagnetic wave

4. Build a timeline of the stages in the physics development:

1. 4th century BC ….

5. Make your own top list of the most important discoveries and inventions of all times. Explain your choice. Use information from the text, your own knowledge and give a short talk about some famous physicists. Don't give their names. The rest of the group will have to guess the names of the scientists.

6. Mark the statements T for 'true' or F for 'false'. Correct the false ones and expand on the true ones.

1. As an independent science physics dates back to the deepest antiquity. 2. Astronomical observations did not influence the development of mathematics. 3. Aristotle thought of experimentation as an important part of any research process. 4. It was Nicolaus Copernicus who suggested that the planets move around the Sun. 5. German astronomer Johannes Kepler came to the conclusion that the planets follow circular orbits. 6. Galileo constructed the first telescope in the world and with its help observed the phases of the planet Jupiter. 7. Isaac Newton founded classical mechanics. 8. James Maxwell was the first who introduced the theory of electromagnetism. 9. The development of nuclear physics re-

sulted in the discovery of X-rays. 10. M. Planck suggested the quantum nature of energy.

7. Answer the questions:

1. How did the ancients explain natural phenomena?
2. What, according to Aristotle, was the key to discovering the laws that govern natural phenomena?
3. Who contributed to the start of the scientific revolution in Europe?
4. What did Kepler correct in the Copernican heliocentric system?
5. What discoveries foreshadowed the work of I. Newton?
6. What ideas did James Maxwell bring together?
7. Who confirmed Maxwell's equations?
8. What scientists worked in the field of thermodynamics?
9. What discoveries resulted in the science of nuclear physics?
10. Who was the originator of quantum mechanics?

8. Give Russian equivalents for the following expressions:

1. Origins date back to the deepest antiquity, 2. could not but affect, 3. have nothing to do with, 4. reveal the importance of observation, 5. conduct experimentation, 6. develop the principles of equilibrium states and the centers of gravity, 7. introduce the heliocentric system, 8. follow not circular but elliptical orbits, 9. by observing the phases, 10. foreshadow the work, 11. set out, 12. in proving, 13. the starting point, 14. minute and discrete quantized amounts, 15. take for granted.

9. Match the words in italics with their correct definitions:

a. the part of mathematics that deals with changing quantities **b**. to act as a warning or sign of a future event **c**. the ability of the object to flow **d**. the force that makes things fall to the ground **e**. having the sun at the centre, with the Earth and other planets moving around it **f**. extremely small **g**. the smallest quantity of radiant energy **h**. a mathematical statement in which you show that two amounts are equal using mathematical symbols **i**. an attempt to find out the truth about something **j**. used to describe the particular way in which something started to exist **k**. a situation in which the sun is covered by the moon **l**. a state of balance **m.** an intelligent and well-educated person **n**. shaped like a circle **o**. the central part of an atom, usually made up of protons and neutrons

Text 2. Nicolaus Copernicus

10. Listen to a radio programme about Copernicus. Then listen again and complete these notes. https://www.onestopenglish.com/download?ac=4682

The Sun went round the **1**_____. Nicolaus Copernicus provided the first **2**_____ theory of the solar system, studied in many fields, including **3**_____and astronomy. His ideas lead the way for science to **4**_____theories already held. Firstly, the universe doesn't have one **5**_____. Secondly, the Earth is not the centre of the **6**_____. Thirdly, any centre of the universe is near the Sun. Fourthly, the distance from the Earth to the **7**_____is nothing if you compare it to the distance from the Earth to the **8**_____. He gave the correct order of the known **9**_____ and explained the phenomenon of **10**_____.

Text 3. Isaak Newton

11. Read the text. Fill in the blanks with the proper words from the box.

contribution velocity masterpiece force theory acceleration motion mathematician phenomena knight discoverer foremost

Sir Isaac Newton (1642–1727), English **1**_____ and physicist, one of the **2**_____ scientific intellects of all time, was born in Lincolnshire. He entered Cambridge University and in two years he already made a number of important contributions to mathematics, optics, mechanics and astronomy.

We owe to Newton our present understanding of motion. He understood clearly the relationships among position, displacement, **3**_____, and **4**_____. Newton is the **5**_____ of the Law of Gravitation. Gradually he came to the conclusion that the **6**_____ which pulled the apple to the ground was the same as the force which kept the Moon to its orbit around the Earth. He extended his **7**_____ of gravitation to the movements of the planets round the Sun. Among fundamentals to mechanics are the three laws of **8**_____ formulated by I. Newton. He summarized his theories in his **9**_____ «The Principia» which was published in 1687. He could unite various **10**_____ of the universe by means of his Universal Law of Gravitation.

Newton was highly appreciated by scientists both at home and abroad. In 1703 he was elected President of the Royal Society. In 1705 he was given the title of **11**_____ for his great **12**_____ to world science.

12. Make a summary of the text using the following phrases:

1.The title of the text is… 2. The text is about… The text deals with… 3. The text covers such points as… first… second… third… 4. It should be underlined that… 5. In conclusion, … 6. To my mind… In my opinion…

13. Go online and find information about a physicist you really admire. Write an essay of 150-200 words about this physicist. Make use of the guidelines:

family background, education, personal qualities and interests, professional career, most outstanding scientific achievements

Text 4. The main branches of physics

Physics, the most fundamental physical science, is concerned with the fundamental principles of the Universe. It is the foundation upon which the other sciences – astronomy, biology, chemistry, and geology – are based. Physics is divided into two main parts: Classical Physics and Modern Physics. Physics up to 1900 AD is considered as Classical physics. It primarily deals with different laws of motion and gravitation. Modern Physics is primarily centered around the two theories of relativity and quantum mechanics.

14. Match the main branches of physics and the areas they deal with:

Classical Mechanics	a. a theory describing objects moving at any speed, even speeds approaching the speed of light
Electrodynamics	b. deals with heat, work, temperature, and the statistical behavior of systems with large numbers of particles
Optics	c. concerns the motion of objects that are large relative to atoms and move at speeds much slower than the speed of light
Thermodynamics	d. a collection of theories connecting the behavior of matter at the submicroscopic level to macroscopic observations
Relativity	e. the study of the behavior of light and its interaction with materials
Quantum mechanics	f. deals with electric and magnetic phenomena associated with charged and magnetic bodies

Text 6. Classical and modern physics

15. Read the text. Fill in the blanks with the given verbs in the correct form:

represent occur lay constitute found last originate develop solve introduce revolutionize distinguish

Until the early 17th century there was a period of accumulation of physical knowledge about natural phenomena. Physics as a science **1**____ from G. Galileo the founder of the exact natural science. The

period from G. Galileo to I. Newton **2**_____ the initial phase of physics, the period of its formation.

The next period begins with the works of I. Newton, who developed the first physical structure of the world as a complete system of mechanics. Physicists **3**_____ classical physics, which was mostly developed before 1895, from modern physics, which is based on discoveries made after 1895. The great system of classical physics **4**_____ two centuries but faced a lot of limitations and serious crises that couldn't be **5**_____ using the physics laws of that time.

Monumental changes **6**_____ at the late 19th century. Maxwell's theory of electromagnetic field challenged Newtonian physics, the further development of which led to revolutionary changes. It was a transition period to new, modern physics, the foundation of which was **7**_____ by a special theory of relativity and quantum theory. In 1905 A. Einstein **8**_____ the special theory of relativity and M. Planck **9**_____ the notion of light quantum and **10**_____ quantum theory.

Quantum theory **11**_____ our understanding of atomic and subatomic processes, just as Albert Einstein's theories of relativity revolutionized our understanding of gravity, space, and time. Together these theories laid the foundation of a new physical quantum-relativistic structure of the world and **12**_____ the most spectacular breakthroughs of twentieth-century physics.

16. Find the following collocations in the text:

1. начало 17 века/конец 19 века 2. накопление физических знаний 3. основатель точного естествознания 4. период становления 5. несмотря на достижения классической физики 5. создать первую физическую картину мира 6. завершенная система механики 7. бросить вызов физике Ньютона 8. привести к революционным изменениям в физике 9. отличать классическую физику от современной 10. переходный период к новой, современной физике 11. специальная теория относительности 12. квантовая теория 13. заложить фундамент 14. квантово-релятивистская структура мира 15. представлять собой самые впечатляющие прорывы

17. Read the text again and summarize it in English using the word combinations in Task 16.

18. Work in teams. For each field of physics (1-10) brainstorm two or three terms that go with it. Match the field of physics with the area(s) of its application:

1. Optics 2. Biophysics 3. Radiophysics 4. Nuclear physics 5. Nanophysics 6. Condensed matter physics 7. Astrophysics 8. Particle physics 9. Acoustics 10. Mechanics

a. to create large capacity disks b. to develop medical imaging instrumentation c. to make new materials d. to set up satellite communication e. to build telescopes f. to operate a nuclear reactor g. to produce computer chips h. to design and create smart machines i. to modify microorganisms for biofuel and bioelectricity j. to develop atomic size machines k. to determine the age of an ancient object or a person l. to create better concert halls m. to develop lasers n. to understand the birth and evolution of the Universe o. to develop intercontinental broadband data channels p. to examine the level of safety of the car and its occupants

19. Read the definition of the word 'idiom' and match the idioms to their meanings:

Idiom is a group of words established by usage as having a meaning not deducible from those of the individual words.

1. blow a fuse 2. get / have our wires crossed 3. not rocket science 4. on the same wavelength 5. reinvent the wheel 6. up and running 7. hit the panic button

a. ready to use b. to react rapidly or by taking emergency precautions c. to get angry d. to think similarly or to have some ideas and opinion e. something is easy to understand or do f. to waste time doing something that has already been done effectively g. to misunderstand someone

Text 7. Physics in the 20th century

20. Read the text and match the English collocations with Russian equivalents:

The past twentieth century was the century of physics and, above all, of quantum physics. First of all, this is the discovery of uranium *fission*, a number of discoveries in *semiconductor* physics and the invention of the *laser*.

The discovery of uranium fission which would serve as the basis of nuclear power engineering played a huge role in the history of the twentieth century. Other important events of the twentieth century were the theory of *superconductivity* and the invention of the *transistor*. Today, lasers have received the most diverse applications – technological, medical, military. One of the most important applications in the information sphere has been the creation of lasers based on semiconductor *heterojunctions*. *Single crystal* laser composed of various elements has become the heart of *fiber-optic* communication.

It is believed that quantum physics of a solid state is today the basis for further scientific and technological progress. One of the most modern devices based on *nanostructures* are lasers and *quantum dot* transis-

tors. Another important example of physics giving rise to technology is the silicon 'chip' that *triggered* the computer revolution in the last three decades of the twentieth century.

21. Give English equivalents to the following phrases:

a. сверхпроводимость b. лазер на монокристалле c. полупроводниковый гетеропереход d. микросхема e. основа для дальнейшего научно-технического развития f. квантовая точка g. физика полупроводников h. основной компонент оптоволоконной связи i. деление урана j. ядерная энергетика

22. Match the words in italics with their correct definitions:

a. a light-emitting nanocrystal made of semiconductor materials, being used now in TV displays **b**. crystal doped with various impurities **c.** a substance used in electronics whose ability to conduct electricity increases with greater heat **d**. a small electrical device containing a semiconductor, used in televisions, radios, etc. to control or increase an electric current **e**. the use of very thin glass or plastic threads through which light can travel to carry information, especially in phone, television, and computer systems **f**. the ability of certain metals to allow electricity to pass through them without any resistance at very low temperatures. **g**. the process of dividing the nucleus of an atom, resulting in the release of a large amount of energy **h**. a structure of intermediate size between microscopic and molecular structures **i**. an electrical junction between two different materials (as semiconductors) **j**. a device that produces a powerful, highly controlled, narrow beam of light **k**. the property of allowing an electrical current to flow through at a very low temperature

23. Choose one of the discoveries or inventions of modern physics. Write a paragraph about it. Give at least three reasons to prove its significance. Make use of the expressions below:

It made it possible to …/It helped to… It was a breakthrough in… It made a revolution in…It made an important contribution to… It gave rise to…/It gave birth to…It laid the foundation for…It found widespread application in…

Text 8. Link between technology and physics

24. Read the text. Fill in the blanks with the given verbs in the correct form:

root arise touch usher kick-started operate
dominate derive develop treat conceive gain

Did you know that you already love physics? Consider the technology around you. Your computer's microchips were built from transistors

using quantum physics. GPS navigation? Sensitive to nanosecond calculations **1**_____ from Einstein's general theory of relativity. PET scans use 3D body-imaging made possible by particle physics. Even the World Wide Web was a creation of physics – **2**_____ by Tim Berners-Lee at CERN to transfer large physics data files.

Today's technology is so deep **3**_____ in physics that our way of life depends upon it. The increased computational power and higher standard of living **4**_____ from physics-based technology allows us to **5**_____ a deeper understanding of physics, again leading to improved technology – and so on.

The connection between physics, technology and society can be seen in many examples. The discipline of thermodynamics **6**_____ from the need to understand and improve the working of steam engines which subsequently **7**_____ the age of industrialization.

The contributions that physicists had on the commercialization of electricity is undoubtedly exceptional. They **8**_____ what is today called the age of globalization. Even the gradual shift to alternative and renewable sources of energy is being influenced by physics. Dams, solar panels, wind farms, nuclear reactors – the energy of the future is overwhelmingly **9**_____ and influenced by the physics of today. In the communications industry, which has connected the world digitally both telephones and televisions **10**_____ on the principle of electromagnetic waves. In the medical industry X-rays are today the most conclusive, and inexpensive method of determining fractures in the body. Nuclear medicine today includes therapies that use radioactive elements to **11**_____ conditions like hyperthyroidism and certain types of lymphoma.

Thus, physics **12**_____ all areas of lives, illuminates them all and helps improve the quality of our life.

25. Match the technology and the scientific principle underlying it. What other examples can be suggested?

Technology	**Scientific principle(s)**
Steam engine	Laws of thermodynamics
Nuclear reactor	Controlled nuclear fission
Radio and Television	Generation, propagation and detection of electromagnetic waves
Computers	Digital logic
Lasers	Light amplification by stimulated emission of radiation
Rocket propulsion	Newton's laws of motion

Electric generator	Faraday's laws of electromagnetic induction
Hydroelectric power	Conversion of gravitational potential energy into electrical energy
Aeroplane	Bernoulli's principle in fluid dynamics
Particle accelerators	Motion of charged particles in electromagnetic field
Sonar	Reflection of ultrasonic waves
Optical fibres	Total internal reflection of light
Electron microscope	Wave nature of electrons
Photocell	Photoelectric effect
Giant Metrewave Radio Telescope (GMRT)	Detection of cosmic radio waves

Text 9. The endless Universe?

Cosmology is the study of the universe as a whole, including theories about its origin, evolution, large-scale structure, and future. Model of an early stage of an expanding universe: the universe is homogeneous. Model of a later stage of an expanding universe: new galaxies emerge continually within the expanding space.

Search the Internet and/or popular science magazines to find out more about current studies of the Universe. Choose the one you find most interesting and make a five-minute PowerPoint presentation about it. Use these questions as the guidelines.

What is being investigated? Who is doing the research? How long has it been under way? What is the purpose and the novelty of the research?

The Big Bang Theory

'So, if you feel you are in a black hole don't give up there's a way out!'
Stephen Hawking

26. Read the text and match the words from the text with their definitions.

Stephen Hawking, one of the world's most brilliant minds, made it possible to understand the universe's *evolution* and what makes up the *universe*. Hawking mathematically proved that *black holes* originate when stars collapse under their own gravitational force.

In the first moments after the Big Bang, the universe was extremely hot and *dense*. As the universe cooled, conditions became just right to give rise to the building blocks of matter – the *quarks* and electrons. A

few millionths of a second later, quarks aggregated to produce *protons* and *neutrons*. Within minutes, these protons and neutrons combined into *nuclei*. Present observations suggest that the first stars formed from clouds of gas around 150–200 million years after the Big Bang. Heavier atoms such as carbon, oxygen and iron, have since been continuously produced in the hearts of stars and catapulted throughout the universe in spectacular stellar explosions called *supernovae*.

Astronomical and physical calculations suggest that a very large fraction of the universe, is made of an unknown type of matter called «*dark matter*». Unlike stars and galaxies, dark matter does not emit any light or electromagnetic radiation, so that we can detect it only through its gravitational effects.

a. any of several elementary particles that are postulated to come in pairs of similar mass **b**. nonluminous matter not yet directly detected by astronomers **c**. everything that exists, including stars, space, etc. **d**. an uncharged elementary particle that has a mass nearly equal to that of the proton **e**. any process of formation or growth; development **f**. the explosion of a star in which the star may reach a maximum intrinsic luminosity **g**. the positively charged central portion of an atom that comprises nearly all of the atomic mass and that consists of protons and usually neutrons **h**. an elementary particle that along with the neutron is a constituent of all other atomic nuclei, that carries a positive charge numerically equal to the charge of an electron **i**. having little space between items or parts **j**. a celestial object that has a gravitational field so strong that light cannot escape it

27. Give Russian equivalents for the following expressions:

1) makes up the universe; 2) as the universe cooled; 3) quarks aggregated; 4) in the hearts of stars; 5) catapulted throughout; 6) spectacular stellar explosions; 7) large fraction; 8) unlike; 9) emit light or electromagnetic radiation

28. Listen and fill in missing words. https://youtu.be/D6lFGJdwRyo

1. Nothing, even light can escape Black Holes _____. 2. In the centre of a black hole, there is something called a _____. 3. _____ is squashed into such a small space that the force of gravity becomes _____. 4. Everything is crushed into a point of infinite _____, punching a hole through the fabric of the universe, and tearing up the rulebook of physics. 5. Vacuum is fizzing with particles and antiparticles that pop into existence from _____ and then disappear. 6. This _____ of escaping particles is known as, «Hawking radiation». 7. Particles that fell into the black hole have a negative _____, and cause the black hole to get small-

er, and smaller, until it disappears. 8. In its final moments a black hole will _____ with the energy of a million nuclear bombs. 9. Stephen Hawking showed that at one point everything in our universe was squeezed into a singularity, which then exploded into the Big Bang, eventually forming _____, stars, planets. 10. He came up with all these profound _____ without the convenience of being able to write anything down. He did it all by thought alone.

29. **Listen to S. Hawking speaking on this issue:** https://youtu.be/gU6yHXJuowU

REVISION

1. Слово «физика» *происходит* от греческого слова, означающего *природу*. Физика от древнейших времен до *начала 17 века* – это период *накопления знаний* об отдельных *явлениях природы*. Ученые наблюдали, что небесные тела двигаются по *кругу*.

2. Аристотель открыл важность *наблюдения*, Архимед *разработал* принципы состояний *равновесия* и центров *тяжести* и *раскрыл* принцип *плавучести*, Кеплер сформулировал три *закона* планетарного *движения*, Н. Коперник *создал гелиоцентрическую модель солнечной* системы.

3. Физика как наука *берет начало* от Г. Галилея – *основоположника точного естествознания*. И. Ньютон *создал* первую *физическую картину мира* как *завершенную систему механики*. Развитие *теории электромагнитного поля* Максвелла *привело к революционным изменениям* в физике. *Классической физика просуществовала* два века, и только *в конце 19 века* начался переход к *современной физике*, фундамент которой *заложили специальная теория относительности* А. Эйнштейна и *квантовая теория* М. Планка.

4. Открытие *деления урана*, создание *теории сверхпроводимости*, изобретение *транзистора* и лазера сыграли огромную роль в истории 20 столетия. *Лазер – монокристалл*, состоящий из различных элементов, стал сердцем *волоконно-оптической связи*. Современные приборы на основе *наноструктур* это лазеры и транзисторы на *квантовых точках*. Создание кремниевых чипов *вызвало компьютерную революцию* в последние три десятилетия двадцатого века. *Современные технологии* настолько глубоко *укоренились* в физике, что от них зависит наш образ жизни.

5. Что *составляет Вселенную*? С. Хокинг *математически доказал*, что черные дыры *возникают*, когда *звезды коллапсируют под действием собственной гравитационной силы*. Строительные бло-

ки материи — *кварки объединяются*, чтобы произвести *протоны* и *нейтроны*, которые в свою очередь *объединяются* в *ядра*. Более тяжелые атомы, такие как углерод, кислород и железо, непрерывно образуются в недрах звезд и катапультируются по всей Вселенной в результате *зрелищных звездных взрывов*, называемых *сверхновыми*.

UNIT 3. MATHEMATICS

Lead-in:

1. What springs to mind when you hear the word 'mathematics'?
2. How many different branches of mathematics can you name?
3. What is the importance of mathematics in our daily life?
4. Why is Maths called the language of science?
5. What are the pros and cons of being a mathematician?

Text 1. What is mathematics?

1. Match the words with their correct definition:

1. quantity	a) to say numbers one after the other in order, or to calculate the number of people or things in a group
2. number	b) the amount or number of something, especially that can be measured
3. definition	c) to reach an answer or a decision by thinking carefully about the known facts
4. set	d) a formal statement or principle in mathematics, science, etc., from which other statements can be obtained
5. space	e) a statement that explains the meaning of a word or phrase
6. to count	f) the mathematical model for a collection of different things; it contains elements or members, which can be mathematical objects of any kind: numbers, symbols, points in space, lines, other geometrical shapes, variables etc.
7. axiom	g) a sign or symbol representing a unit that forms part of the system of counting and calculating
8. theorem	h) a formal statement that can be shown to be true by logic
9. to measure	i) all of the area in which everything exists, and in which everything has position or direction
10. to deduce	j) to discover the exact size or amount of something
11. property	k) an attribute, characteristic, or quality

2. Read the text. Fill in the gaps with the appropriate words from ex.1. Change the form of the word if necessary.

Mathematics is the product of many lands and it belongs to the whole of mankind. We know how necessary it was even for the early people to learn **1) ...** and to become familiar with mathematical ideas, processes and facts. In the course of time, counting led to arithmetic and **2) ...** led to geometry. Arithmetic is the study of **3) ...**, while geometry is the

study of shape, size and position. These two subjects are regarded as the foundations of mathematics.

It is impossible to give a concise **4) ...** of mathematics as it is a multifield subject. Mathematics in the broad sense of the word is a peculiar form of the general process of human cognition of the real world. It deals with the space forms and quantity relations abstracted from the physical world.

Contemporary mathematics is a mixture of much that is very old and still important (e. g., counting, the Pythagorean theorem) with new concepts such as sets, axiomatics, structure. The totality of all abstract mathematical sciences is called Pure Mathematics. The totality of all concrete interpretations is called Applied Mathematics. Together they constitute Mathematics as a science.

One of the modern **5)** … of mathematics runs as follows: mathematics is the study of relationships among quantities, magnitudes, and **6) ...** of logical operations by which unknown quantities, magnitudes and properties may be **7) ...** .

In the past, mathematics was regarded as the science of quantity, whether of magnitudes, as in geometry, or of numbers, as in arithmetic, or the generalization of these two fields, as in algebra. Toward the middle of the 19th century, however, mathematics came to be regarded increasingly as the science of relations, or as the science that draws necessary conclusions. The latter view encompasses mathematical or symbolic logic, the science of using symbols to provide an exact theory of logical deduction and inference based on definitions, **8) ...,** postulates, and rules for combining and transforming positive elements into more complex relations and theorems.

3. Match the given words and phrases with their Russian equivalents. Memorize. Use them in your own sentences.

A: 1. measuring, 2. to deal with, 3. applied, 4. pure, 5. concept, 6. to transform, 7. to constitute, 8. sets, 9. quantity, 10. lead to

B: a) приводить к, b) измерение (действие), c) прикладной, d) множества, e) понятие, f) теоретический, чистый, h) количество, величина, i) преобразовывать, j)иметь дело с, k)составлять

4. Fill in the blanks with the proper words from the box.

geometry, conclusions, set, applied, to constitute, to deal with, calculations, to deduce, relationship,to lead to

1… studies spatial relationships and shapes, while ignoring other properties of real bodies (density, weight, colour, etc.).2. The smallest mistake in calculation can **...** erroneous results. 3. The Greeks insist-

ed that all math … should be established by deductive reasoning only. 4. A … may have a finite number of elements or be infinite. 5. … mathematics is the application of mathematical methods by different fields such as physics, engineering, medicine, biology, finance, business, computer science, and industry. 6. Discrete mathematics and algorithms … a lingua franca for computer scientists and software developers. 7. Mathematics is the branch of science that … numbers, involves … and mainly focuses on the study of quantity, shapes, measurements etc. 8. The objective of maths activity consists of the theorems … from a set of axioms.9. The concept of a function, or a … between variables, is almost totally a mental creation.

5. Answer the questions.

1. What did counting and measuring lead to? 2. What is mathematics in the broad sense of the word? 3. What does it deal with? 4. What is Pure Mathematics? 5. How is Applied Mathematics defined? 6. What is one of the modern definitions of mathematics? 7. How was mathematics interpreted in the past? 8. What is it considered to be now?

6. Retell the text.

Text 2. Main branches of mathematics

1. Read the text and follow the tasks below the text:

1. Arithmetic. This is one of the most basic branches of mathematics. Arithmetic deals with numbers and their applications in many ways. Addition, subtraction, multiplication, and division form its basic groundwork as they are used to solve a large number of questions and progress into more complex concepts like exponents, limits, and many other types of calculations. This is one of the most important branches because its fundamentals are used in everyday life for a variety of reasons from simple calculations to profit and loss computation.

2. Algebra. A broad field of mathematics, algebra deals with solving generic algebraic expressions and manipulating them to arrive at results. Unknown quantities denoted by alphabets that form a part of an equation are solved for and the value of the variable is determined. A fascinating branch of mathematics, it involves complicated solutions and formulas to derive answers to the problems posed.

3. Geometry. Dealing with the shape, sizes, and volumes of figures, geometry is a practical branch of mathematics that focuses on the study of polygons, shapes, and geometric objects in both two-dimensions and three-dimensions. Congruence of objects is studied at the same time focusing on their special properties and calculation of their area, volume,

and perimeter. The importance of geometry lies in its actual usage while creating objects in practical life.

4. Trigonometry. Derived from Greek words «trigonon» meaning triangle and «metron» meaning «measure», trigonometry focuses on studying angles and sides of triangles to measure the distance and length. Amongst the prominent branches of mathematics used in the world of technology and science to develop objects, trigonometry is a study of the correlation between the angles and sides of the triangle. It is all about different triangles and their properties!

5. Calculus. It is one of the advanced branches of mathematics and studies the rate of change. With the advent of calculus, a revolutionary change was brought about in the study of maths. Earlier maths could only work on static objects but with calculus, mathematical principles began to be applied to objects in motion. Used in a multitude of fields, the branch can be further categorized into the differential and integral calculus both starkly different from each other. **Differential calculus** deals with the rate of change of a variable and it is a means of finding tangents to curves. **Integral calculus** is concerned with the limiting values of differentials and is a means of determining length, volume, or area.

6. Probability and Statistics. The abstract branch of mathematics, probability and statistics use mathematical concepts to predict events that are likely to happen and organize, analyze, and interpret a collection of data. Amongst the relatively newer branches of mathematics, it has become indispensable because of its use in both natural and social sciences. The scope of this branch involves studying the laws and principles governing numerical data and random events. Presenting an interesting study, statistics, and probability is a branch full of surprises.

7. Number Theory. It is a branch of pure mathematics devoted primarily to the study of integers and integer-valued functions. Number theorists study prime numbers as well as the properties of mathematical objects made out of integers (for example, rational numbers) or defined as generalizations of the integers (for example, algebraic integers). The basic level of Number Theory includes introduction to properties of integers like addition, subtraction, multiplication, modulus and builds up to complex systems like cryptography, game theory and more.

8. Topology. Topology is a much recent addition into the branches of Mathematics list. It is concerned with the deformations in different geometrical shapes under stretching, crumpling, twisting and bedding. Deformations like cutting and tearing are not included in topologies.

Its application can be observed in differentiable equations, dynamical systems, knot theory, and Riemann surfaces in complex analysis.

9. Applied Mathematics. Applied mathematics is the application of mathematical methods by different fields such as physics, engineering, medicine, biology, finance, business, computer science, and industry. Thus, applied mathematics is a combination of mathematical science and specialized knowledge. The term «applied mathematics» also describes the professional specialty in which mathematicians work on practical problems by formulating and studying mathematical models.

2. Give the Russian and English equivalents:

A) 1. simple calculations, 2. congruence of objects, 3. the rate of change, 4. to find tangents to curves, 5. limiting values of differentials, 6.to predict events, 7. integer-valued functions, 8. differential equations, 9. Riemann surface, 10. area, 11. random events, 12.to determine the value of the variable, 13. the correlation between, 14. exponent, 15.to arrive at results, 16. prime numbers, 17. modulus, 18. denoted by alphabets

B) 1. сложение, 2. вычитание, 3. умножение, 4. деление, 5. неизвестная величина, 6. измерять расстояние и длину, 7. дифференциальное исчисление ,8. интегральное исчисление ,9. целое число, 10. геометрические фигуры ,11. теория узла, 12. решать алгебраические выражения, 13. уравнение, 14. угол, 15. треугольник, 16. многоугольник.

3. Fill in the blanks with the proper words from the box:

Equation, polygon, tangent, exponent, congruence, determine, expression, quantity, angle, subtraction,integer, primer

1… is an operation used to find the difference between numbers. 2. A closed two-dimensional figure with three or more straight lines is called a … . 3. A mathematical statement in which you show that two amounts are equal using mathematical symbols is an … 4. In geometry, an … is the space between 2 rays (or line segments) with the same endpoint (or vertex). 5. In algebra, a variable is used to represent an unknown … or unknown values. 6. An … is a number with no decimal or fractional part and it includes negative and positive numbers, including zero. 7. A … number is a number that can only be divided by itself and 1 without remainders. 8. An algebraic … is a series of numbers and variables combined by mathematical operations (addition, subtraction, multiplication, etc.).9, How can I use the following formula to … the value of a variable?10. … means two objects (whether two dimensional or three dimensional) are identical in size and shape. 11. A **...** to a curve

is a line that touches the curve at one point and has the same slope as the curve at that point.12, An **...** is the number of times that a certain number is multiplied by itself. It is written as a small-sized number at the top right of the number.

4. Match the terms with their definitions.

Arithmetic	a.deals with the rate of change of a variable and it is a means of finding tangents to curves
Algebra	b. use mathematical concepts to predict events that are likely to happen and organize, analyze, and interpret a collection of data
Geometry	c. is concerned with the deformations in different geometrical shapes under stretching, crumpling, twisting and bedding
Differential calculus	d.deals with numbers and their applications in many ways
Integral calculus	e. is the application of mathematical methods by different fields such as physics, engineering, medicine, biology, finance, business, computer science, and industry
Probability and statistics	f. is devoted primarily to the study of integers and integer-valued functions
Number Theory	g. focuses on the study of polygons, shapes, and geometric objects in both two-dimensions and three dimensions
Topology	h. studies entirely abstract concepts
Pure mathematics	i.deals with solving generic algebraic expressions and manipulating them to arrive at results
Applied mathematics	j. is concerned with the limiting values of differentials and is a means of determining length, volume, or area

5. Answer the following questions:

1. What are the specific branches of mathematics? 2. What does arithmetic deal with? 3. What does algebra involve? 4. Why is geometry called a practical branch of mathematics? 5. What objects does trigonometry study? 6. Why did calculus bring a revolutionary change in the study of maths? 7. What do differential calculus and integral calculus deal with? 8. Why are probability and statistics called indispensable? 9. What is Number Theory devoted to? 10. What does the basic level of Number Theory include? 11. What is topology concerned with? 12. What does applied maths deal with?

6. Speak about the main branches of mathematics.

Text 3. Development of Mathematics in the 17th–19th centuries

1. Read and translate the text.

The scientific revolution of the 17th century spurred advances in mathematics as well. The founders of modern science – **Nicolaus Copernicus**, **Johannes Kepler**, **Galileo**, and **Isaac Newton** – studied the natural world as mathematicians, and they looked for its mathematical laws. Over time, mathematics grew more and more abstract as mathematicians sought to establish the foundations of their fields in logic.

The 17th century opened with the discovery of logarithms by the Scottish mathematician **John Napier** and the Swiss mathematician **Justus Byrgius**. Logarithms enabled mathematicians to extract the roots of numbers and simplified many calculations by basing them on addition and subtraction rather than on multiplication and division.

Napier, who was interested in simplification, studied the systems of the Indian and Islamic worlds and spent years producing the tables of logarithms that he published in 1614. **Kepler**'s enthusiasm for the tables ensured their rapid spread.

The 17th century saw the greatest advances in mathematics since the time of ancient Greece. The major invention of the century was calculus. Although two great thinkers – Sir Isaac Newton of England and **Gottfried Wilhelm Leibniz** of Germany – have received credit for the invention, they built on the work of others. As Newton noted, «If I have seen further, it is by standing on the shoulders of giants.» Major advances were also made in numerical calculation and geometry.

Gottfried Leibniz was born (1st July, 1646) and lived most of his life in Germany. His greatest achievement was the invention of integral and differential calculus, the system of notation which is still in use today. In England, Isaac Newton claimed the distinction and accused Leibniz of plagiarism, that is stealing somebody else's ideas but stating that they are original. Modern-day historians, however, regard Leibniz as having arrived at his conclusions independently of Newton. They point out that there are important differences in the writings of both men.

Differential calculus came out of problems of finding tangents to curves, and an account of the method is published in **Isaac Barrow**'s «Lectiones geometricae» (1670). Newton had discovered the method (1665-66) and suggested that Barrow include it in his book.

Leibniz had also discovered the method by 1676, publishing it in 1684. Newton did not publish his results until 1687. A bitter dispute arose over the priority for the discovery. In fact, it is now known that the two made their discoveries independently and that Newton had made it ten years

before Leibniz, although Leibniz published first. The modern notation of dy/dx and the elongated s for integration are due to Leibniz.

The most important development in geometry during the 17th century was the discovery of analytic geometry by **Rene Descartes** and **Pierre de Fermat**, working independently in France. Analytic geometry makes it possible to study geometric figures using algebraic equations. By using algebra, Descartes managed to overcome the limitations of Euclidean geometry. That resulted in the reversal of the historical roles of geometry and algebra.

The French mathematician **Joseph Louis Lagrange** observed in the 18th century, «As long as algebra and geometry proceeded along separate paths, their advance was slow and their applications limited. But when these sciences joined company, they drew from each other fresh vitality and thenceforward marched on at a rapid pace toward perfection.»

Descartes' publications provided the basis for Newton's mathematical work later in the century. Pierre de Fermat, however, regarded his own work on what became known as analytic geometry as a reformulation of Appollonius's treatise on conic sections. That treatise had provided the basic work on the geometry of curves from ancient times until Descartes.

During the 18th century, calculus became the cornerstone of mathematical analysis on the European continent. Mathematicians applied the discovery to a variety of problems in physics, astronomy, and engineering. In the course of doing so, they also created new areas of mathematics.

In France, Joseph Louis Lagrange made substantial contributions in all fields of pure mathematics, including differential equations, the calculus of variations, probability theory, and the theory of equations. In addition, Lagrange put his mathematical skills to work in the solution of practical problems in mechanics and astronomy.

The greatest mathematician of the 18th century, **Leonard Euler** of Switzerland, wrote works that covered the entire fields of pure and applied mathematics. He wrote major works on mechanics that preceded Lagrange's work. He won a number of prizes for his work on the orbits of comets and planets, the field known as celestial mechanics. But Euler is best known for his works in pure mathematics. In one of his works, Introduction to the Analysis of Infinites, published in 1748, he approached calculus in terms of functions rather than the geometry of curves. Other works by Euler contributed to number theory and differential geometry (the application of differential calculus to the study of the properties of curves and curved spaces).

Mathematicians succeeded in firming the foundations of analysis and discovered the existence of additional geometries and algebras and more than one kind of infinity.

The 19th century began with the German mathematician **Carl Frederich**

Gauss. He ranks as one of the greatest mathematicians of the world. His book Inquiries into Arithmetic published in 1801 marks the beginning of modern era in number theory.

Gauss called mathematics the queen of sciences and number theory the queen of mathematics. Almost from the introduction of calculus, efforts had been made to supply a rigorous foundation for it. Every mathematician made some effort to produce a logical justification for calculus and failed. Although calculus clearly worked in solving problems, mathematicians lacked rigorous proof that explained why it worked. Finally, in 1821, the French mathematician **Augustin Louis Cauchy** established a rigorous foundation for calculus with his theory of limits, a purely arithmetic theory.

Later, mathematicians found Cauchy's formulation still too vague because it did not provide a logical definition of real number. The necessary precision for calculus and mathematical analysis was attained in the 1850s by the German mathematician **Karl T. W. Weierstrass** and his followers.

Another important advance in analysis came from the French mathematician **Jean Baptiste Fourier**, who studied infinite series in which the terms are trigonometric functions. Known today as Fourier series, they are still powerful tools in pure and applied mathematics. The investigation of Fourier series led another German mathematician, **Georg Cantor**, to the study of infinite sets and to the arithmetic of infinite numbers. Georg Cantor began his mathematical investigations in number theory and went on to create set theory. In the course of his early studies of Fourier series, he developed a theory of irrational numbers. Cantor and another German mathematician, **Julius W. R. Dedekind**, defined the irrational numbers and established their properties. These explanations hastened the abandonment of many 19th century mathematical principles. When Cantor introduced his theory of sets, it was attacked as a disease from which mathematics would soon recover. However, it now forms part of the foundations of mathematics. The application of set theory greatly advanced mathematics in the 20th century**.**

2. Give the Russian and English equivalents:

A) 1. cornerstone, 2. substantial, 3. major works, 4. to rank as, 5. to lack smth. 6.

vague, 7. precision,8. to attain, 9. abandonment, 10. to advance, 11. simplification,12.to approach, 13. to elongate, 14. pace, 15. tangents to curves

B) 1. сделать открытие, 2. извлекать корни, 3. упростить, 4. интегральные и дифференциальные исчисления, 5. система обозначений, 6. совершенство, 7. искривленное пространство, 8. теория чисел, 9. достичь, 10. ускорять, 11. теория пределов

3. Match the names of the mathematicians with their contributions to science.

Mathematician	Contribution
1. Georg Cantor	a. infinite series in which the terms are trigonometric functions
2. Julius W. R. Dedekind	b. analytic geometry
3. Augustin Louis Cauchy	c. pure mathematics, including differential equations, the calculus of variations, probability theory, and the theory of equations
4. Jean Baptiste Fourier	d. a theory of irrational numbers
5. Joseph Louis Lagrange	e. tables of logarithms
6. Rene Descartes and Pierre de Fermat	f. integral and differential calculus
7. Gottfried Wilhelm Leibniz	g. irrational numbers and their properties
8. John Napier	h. laws of motion
9. Isaac Newton	i. foundation for calculus with his theory of limits, a purely arithmetic theory

4. Complete the sentences below with the words and phrases from the box.

a. Rene Descartes and Pierre de Fermat b. the discovery of calculus c. Kepler d. preceded Lagrange's work e. Newton and Leibniz f. the scientific revolution of the 17th century g. the tables of logarithms h. physics, astronomy, and engineering	i. celestial mechanics j. Fourier series k. Karl T. W. Weierstrass l. mechanics and astronomy m. number theory and differential geometry n. Carl Frederich Gauss o. Cantor and Dedekind

1. The Scottish mathematician Napier spent years producing ….
2. The rapid spread of the tables of logarithms was ensured by ….
3. The development of analytic geometry was beneficial for both ….

4. The invention of calculus is connected with the names of …. 5. A bitter dispute arose over the priority for …. 6. Advances in mathematics were facilitated by …. 7. Euler's major works on mechanics …. 8. Mathematicians applied the discovery of calculus to …. 9. Lagrange managed to solve some practical problems in …. 10. Euler's works contributed to …. 11. Euler won a number of prizes for his work on …. 12. Mathematics was called the queen of sciences by …. 13. Cantor's study of infinite sets became possible due to the study of …. 14. The properties of irrational numbers were established by …. 15. A precise foundation for calculus was supplied by ….

5. Answer the questions:

1.What scholars are considered to be the founders of modern science? 2. Why did mathematics grow more and more abstract? 3. Who were logarithms discovered by? 4. What did logarithms enable mathematicians to do? 5. What was the major invention of the 17th century? 6. What is the essence of analytic geometry? 7. Why did a dispute arise between Leibniz and Newton? 8. What enabled Descartes to overcome the limitations of Euclidean geometry? 9. Whose publications provided the basis for Newton's mathematical work later in the century? 10. What did the discovery of calculus lead to? 11. What was Lagrange's contribution to pure and applied mathematics?12. What did Euler's works contribute to? 13. What is the essence of differential geometry? 14. What event marked the beginning of modern era in number theory? 15. When was a rigorous foundation for calculus finally supplied? 16. What is the theoretical and practical value of Fourier series? 17. What was Georg Cantor's contribution to mathematical studies? 18. Who were irrational numbers investigated and defined by? 19. What was the first reaction to Cantor's set theory? 20. Was the attitude to the discovery later changed?

6. Speak about 2 mathematicians you admire. Use the facts from *the previous text* and give some extra information.

REVISION

1. Translate from Russian into English:

1. Математика — это наука о *количественных отношениях* и *пространственных формах* действительного мира. Она занимается *числами,* включает в себя *вычисления* и фокусируется на изучении *количества, форм, измерений* и т. д.

2. Разделами математики являются *арифметика, алгебра, геометрия и топология, тригонометрия, математический анализ,*

теория чисел, теория вероятностей и математическая статистика, прикладная математика и др.

3. Алгебра занимается *решением алгебраических выражений* и манипулированием ими для получения результатов.

4. В алгебре *переменная* используется для представления *неизвестной величины или неизвестных значений.*

5. Математической статистикой называется раздел *прикладной математики,* изучающий методы сбора, обработки и анализа *статистических данных* для научных и практических целей. Основными объектами изучения в теории вероятностей являются *случайные события и случайные величины.*

6. Тригонометрия – это математическая дисциплина, изучающая *зависимости между углами и сторонами треугольников* и тригонометрические функции.

7. Использование в вычислениях вместо чисел их *логарифмов* позволяет заменить *умножение* более простой операцией *сложения, деление – вычитанием, возведение в степень – умножением и извлечение корней* – делением. Математики Йост Бюрги и Джон Непер составили *таблицы логарифмов* в 17 веке.

8. Ньютон и Лейбниц считаются создателями *математического анализа.*

9. Георг Кантор известен как создатель *теории множества.*

10. Аналитическая геометрия, созданная Пьером де Ферма и Рене Декартом, позволяет изучать геометрические фигуры с помощью *алгебраических уравнений.*

11. Исаак Бэрроу, учитель Исаака Ньютона, разработал метод определения *касательных к кривым.*

12. Леонард Эйлер – величайший математик в истории, в геометрии положил начало *топологии,* науки об общих *свойствах пространства и фигур.*

13. Иоганн Карл Фридрих Гаусс считается «королем математики», его книга «Исследования по арифметике» (Inquiries into Arithmetic), опубликованная в 1801 году, знаменует начало современной эры в *теории чисел*. Теория чисел – это раздел теоретической («чистой») математики, посвященный изучению *целых чисел и целочисленных функций.*

UNIT 4. CHEMISTRY

Lead-in:

1. What associations do you have when you hear the word 'chemistry'?
2. How many different areas of specialization can you name?
3. Where is chemistry applied in our daily life?
4. Will you agree that chemistry is the central science? Why?
5. What makes studying chemistry important?

Text 1. What is chemistry

1. Study the vocabulary and match the words with their definitions:

Science	: the act or process of composing; specifically : arrangement into specific proportion or relation and especially into artistic form : the arrangement of type for printing : the production of type or typographic characters (as in photocomposition) arranged for printing
Composition	nonmetallic chemical element of Group 16 of the periodic table. It is colourless, odourless, tasteless gas essential to living organisms, being taken up by animals, which convert it to carbon dioxide; plants, in turn, utilize carbon dioxide as a source of carbon and return it to the atmosphere. It forms compounds by reaction with practically any other element, as well as by reactions that displace elements from their combinations with each other; in many cases, these processes are accompanied by the evolution of heat and light and in such cases are called combustions. Its most important compound is water
Coherent	: logically or aesthetically ordered or integrated : consistent : having clarity or intelligibility : understandable : having the quality of holding together or cohering; especially: cohesive, coordinated
Deoxyribonucleic acid (DNA)	any of various self-contained units of matter or energy that are the fundamental constituents of all matter. They include electrons, the negatively charged, almost massless particles that nevertheless account for most of the size of the atom, and they include the heavier building blocks of the small but very dense nucleus of the atom, the positively charged protons and the electrically neutral neutrons

Oxygen (O)	: the basic building block of chemistry. It is the smallest unit into which matter can be divided without the release of electrically charged particles. It also is the smallest unit of matter that has the characteristic properties of a chemical element
Subatomic particle – also called elementary particle	is an organic chemical that contains genetic information and instructions for protein synthesis. It is found in most cells of every organism. It is a key part of reproduction in which genetic heredity occurs through the passing down of one from parent or parents to offspring
An atom	: the capacity for doing work. It may exist in potential, kinetic, thermal, electrical, chemical, nuclear, or other forms. In the International System of Units (SI), energy is measured in joules. One joule is equal to the work done by a one-newton force acting over a one-meter distance
Energy	: any system of knowledge that is concerned with the physical world and its phenomena and that entails unbiased observations and systematic experimentation. In general, a science involves a pursuit of knowledge covering general truths or the operations of fundamental laws
Substance	gather together or acquire an increasing number or quantity of
Accumulate	made larger; enlarged/lasting longer than is usual or expected
Extended	a particular kind of matter with uniform properties

2. Read the text. Fill in the gaps with the words from ex.1. Change the form of the word if necessary.

Chemistry, the **1)** …. that deals with the properties, **2)** …, and structure of substances (defined as elements and compounds), the transformations they undergo, and the **3)** … that is released or absorbed during these processes. Every substance, whether naturally occurring or artificially produced, consists of one or more of the hundred-odd species of **4)** … that have been identified as elements. Although these atoms, in turn, are composed of more elementary particles, they are the basic building blocks of chemical substances; there is no quantity of **5)** …, mercury, or gold, for example, smaller than an atom of that substance. Chemistry, therefore, is concerned not with the **6)** … domain but with the properties of atoms and the laws governing their combinations and how the knowledge of these properties can be used to achieve specific purposes.

The great challenge in chemistry is the development of a **7**) … explanation of the complex behavior of materials, why they appear as they do, what gives them their enduring properties, and how inter-

actions among different substances can bring about the formation of new substances and the destruction of old ones. From the earliest attempts to understand the material world in rational terms, chemists have struggled to develop theories of matter that satisfactorily explain both permanence and change. The ordered assembly of indestructible atoms into small and large molecules, or **8)**… networks of intermingled atoms, is generally accepted as the basis of permanence, while the reorganization of atoms or molecules into different arrangements lies behind theories of change. Thus, chemistry involves the study of the atomic composition and structural architecture of substances, as well as the varied interactions among substances that can lead to sudden, often violent reactions.

Chemistry also is concerned with the utilization of natural substances and the creation of artificial ones. Cooking, fermentation, glass making, and metallurgy are all chemical processes that date from the beginnings of civilization. Today, vinyl, Teflon, liquid crystals, semiconductors, and superconductors represent the fruits of chemical technology. The 20th century saw dramatic advances in the comprehension of the marvelous and complex chemistry of living organisms, and a molecular interpretation of health and disease holds great promise. Modern chemistry, aided by increasingly sophisticated instruments, studies materials as small as single atoms and as large and complex as **9)** …, which contains millions of atoms. New **10)** ... can even be designed to bear desired characteristics and then synthesized. The rate at which chemical knowledge continues to **11)** … is remarkable. Over time more than 8,000,000 different chemical substances, both natural and artificial, have been characterized and produced. The number was less than 500,000 as recently as 1965.

Intimately interconnected with the intellectual challenges of chemistry are those associated with industry. In the mid-19th century the German chemist Justus von Liebig commented that the wealth of a nation could be gauged by the amount of sulfuric acid it produced. This acid, essential to many manufacturing processes, remains today the leading chemical product of industrialized countries. As Liebig recognized, a country that produces large amounts of sulfuric acid is one with a strong chemical industry and a strong economy as a whole. The production, distribution, and utilization of a wide range of chemical products is common to all highly developed nations. In fact, one can say that the «iron age» of civilization is being replaced by a «polymer age» for in some countries the total volume of polymers now produced exceeds that of iron.

3. Match the given words and phrases with their English equivalents:

1. иметь дело с	a. to interconnect
2. претерпевать	b. to occur
3. выделять	c. to undergo
4. поглощать	d. to represent
5. происходить	e. to distribute
6. вовлекать	f. to release
7. представлять	g. to involve
8. взаимодействовать	h. to absorb
9. превышать	i. to deal with
распределять	j. to exceed

4. Fill in the gaps with the words from the box:

science various particular substance chemistry properties molecules undergo concepts requires led

Chemistry has justly been called the central 1) … Chemists study the 2) … substances in the world, with a 3) … focus on the processes by which one 4) … is transformed into another. Today, 5) … is defined as the study of the composition and 6) … of elements and compounds, the structure of their 7) …, and the chemical reactions that they 8) ... Rather than starting with such modern 9) …, though, a fuller appreciation of the subject 10) … an examination of the historical processes that 11) … to these concepts.

5. **Answer the questions**

1. What does chemistry deal with?
2. What is the main challenge in chemistry?
3. What is generally accepted as the basis of permanence?
4. What lies behind theories of change?
5. What does chemistry involve?
6. What represent the fruits of chemical technology?
7. What could the wealth of a nation be gauged by?
8. What remains the leading chemical product of all industrialized countries?

Text 2. Philosophy of matter in antiquity

1. Read and explain the following words

Philosopher [fɪˈlɒsəfə] antiquity [ænˈtɪkwɪtɪ] dozen [dʌzn] substances [ˈsʌbstənsɪz] propound [prəˈpaʊnd] variety [vəˈraɪətɪ] indeterminate [ɪndɪˈtɜːmɪnɪt] irreducible [ɪrɪˈdjuːsəbl] perception [pəˈsepʃn] terrestrial [təˈrestrɪəl] region [ˈriːdʒən] quality [ˈkwɒlɪtɪ] featureless [ˈfiːʧəlɪs] con-

sequently [ˈkɒnsɪkwəntlɪ] precluded [prɪˈkluːdɪd] adjustment [əˈʤʌstmənt] throughout [θruːˈaʊt] metalsmiths [metalsmiths] assayer [əˈseɪə] ceramists [sɪˈræmɪst] dyer [ˈdaɪə] sophisticated [səˈfɪstɪkeɪtɪd] emerged [ɪˈmɜːdʒd] alchemy [alkımi]

2. Read the names and tell the group what they are connected with

Thales of Miletus [teɪlz ɒv maɪˈliːtəs], Anaximander [ə,næksɪ›mændə], Empedocles [ɪm›pedəkliːz], Leucippus [luːˈsɪpəs], Democritus [dɪˈmɒkrɪtəs] Plato [ˈpleɪtəʊ], Pythagoras [paɪˌθæg.ᵊr.əs]

3. Read the article and insert the correct preposition.

of (8) on by (2) from in

Indeed, the philosophers 1) … antiquity could have had no notion that all matter consists 2) … the combinations 3) … a few dozen elements as they are understood today. The earliest critical thinking 4) … the nature 5) … substances, as far as the historical record indicates, was 6) … certain Greek philosophers beginning about 600 BCE. Thales of Miletus, Anaximander, Empedocles, and others propounded theories that the world consisted 7) … varieties 8) … earth, water, air, fire, or indeterminate «seeds» or «unbounded» matter. Leucippus and Democritus propounded a materialistic theory 9) … invisibly tiny irreducible atoms 10) … which the world was made. 11) … the 4th century BCE, Plato, influenced 12) … Pythagoreanism, taught that the world 13) … the senses was but the shadow of a mathematical world of «forms» beyond human perception.

in (2) of (3) on by from (2) for (3) into

14) … contrast, Plato's student Aristotle took the world 15) … the senses seriously. Adopting Empedocles's view that the terrestrial region consisted 16) … earth, water, air, and fire, Aristotle taught that each 17) … these materials were a combination of qualities such as hot, cold, moist, and dry. For Aristotle, these «elements» were not building blocks of matter as they are thought of now; rather, they resulted 18) … the qualities imposed 19) … otherwise featureless prime matter. Consequently, there were many different kinds of earth, 20) … instance, and nothing precluded one element 21) … being transformed 22) … another 23) … appropriate adjustment of its qualities. Thus, Aristotle rejected the speculations of the ancient atomists and their irreducible fundamental particles. His views were highly regarded 24) … late antiquity and remained influential throughout the Middle Ages. 25) … thousands of years before Aristotle, metalsmiths, assayers, ceramists, and dyers

had worked to perfect their crafts using empirically derived knowledge of chemical processes. By Hellenistic and Roman times, their skills were well advanced, and sophisticated ceramics, glasses, dyes, drugs, steels, bronze, brass, alloys of gold and silver, foodstuffs, and many other chemical products were traded. Hellenistic Alexandria in Egypt was a centre 26) … these arts, and it was apparently there that a group of ideas emerged that later became known as alchemy.

4. Translate the sentences from Russian into English.

1. В самом деле, химия – наука, которая занимается свойствами, составом и структурой веществ, превращениями, которым они подвергаются, и энергией, которая выделяется или поглощается во время этих процессов. Каждое вещество, независимо от того, является ли оно природного или искусственного происхождения, состоит из одного или более из ста с лишним видов атомов, которые называются элементами. Хотя эти атомы, в свою очередь, состоят из более простых частиц, они являются основными строительными единицами химических веществ; например, не существует количества кислорода, ртути или золота, меньшего, чем атом этого вещества. Химия, таким образом, имеет дело не с субатомной областью, а со свойствами атомов и законами, управляющими их комбинациями, и с тем, как знание этих свойств может быть использовано для достижения конкретных целей.

2. Химия есть наука об особой форме движения материи, характерной особенностью которой является качественное превращение веществ: в процессе химической реакции одни вещества как бы исчезают, а вместо них появляются новые вещества с новыми свойствами. При этом качественный скачок является результатом как количественного изменения состава вещества (изменение атомарного состава молекул), так и внутренней молекулярной структуры.

3. С самых первых попыток понять материальный мир, используя рациональные термины, химики изо всех сил пытались разработать теории материи, которые бы давали четкое определение как постоянству, так и изменению. Упорядоченное соединение неразрушимых атомов в малые и большие молекулы, или протяженные сети перемешанных атомов, обычно принимаются в качестве основы постоянства, в то время как реорганизация атомов или молекул в различные структуры лежит в основе теорий изменений. Таким образом, химия включает в себя изучение атомного состава и структурной архитектуры веществ, а также разнообразных взаимодействий между веществами, которые могут привести к внезапным, часто бурным реакциям.

Text 2.1. Alchemy

1. Read the text and fill in the gaps with the correct words from the box.

skills origin sophistication fashion sublimating matter transformability combination note

Three different sets of ideas and 1) … fed into the 2) … of alchemy. First was the empirical 3) … of jewelers, gold- and silversmiths, and other artisans who had learned how to 4) … precious and semiprecious materials. Among their skills were smelting, assaying, alloying, gilding, amalgamating, distilling, painting, 5) … and lacquering. The second component was the early Greek theory of 6) …, especially Aristotelian philosophy, which suggested the possibility of unlimited 7) … of one kind of matter into another. The third of alchemy's roots consisted of a complex 8) … of ideas derived from Asian philosophies and religions. Hellenistic mystery religions, and what became known as the Hermetic writings (a body of pseudonymous Greek writings on magic, astrology, and alchemy ascribed to the Egyptian god Thoth or his Greek counterpart Hermes Trismegistos). It is important to 9) …, however, that Hellenistic Egypt is only one of several candidates for the homeland of alchemy; at about the same time, similar ideas were developing in Persia, China, and elsewhere.

manipulate valuable quest transmute craft devoting cure considered immortal spurned achieving

In general, alchemists sought to 10) … the properties of matter in order to prepare more 11) … substances. Their most familiar 12) … was to find the philosopher's stone, a magical substance that would 13) … ordinary metals such as copper, tin, iron, or lead into silver or gold. Important materials in this 14) … included sulfur, mercury, and electrum (a gold-silver alloy). However, many other alchemists spurned alchemical transmutation (aurifaction), 15) … their efforts instead to a pharmaceutical preparation known as the «elixir of life» that would 16) … any disease, including the ultimate disease, death. The philosopher's stone and the elixir of life could be 17) … parallel quests, for each would «cure» metallic or human bodies, respectively, yielding 18) … perfection. There was a parallel religious dimension to all this as well. Finally, some alchemists 19) … material manipulations entirely, devoting themselves to meditation with the goal of 20) … spiritual purity and ultimate redemption.

scholars scientific pursued derived suggests chemistry medieval sulfur mercury altering corpus

After the rise of Islam, Arabic-speaking 21) … of the 9th century translated Greek 22) … and philosophical works into their own language. Thereafter, philosophers in the Islamic world 23) … chemical and alchemical ideas with enthusiasm and success. The sizable number of modern chemical words 24) … from Arabic—*alcohol*, *alkali*, *alchemy*, *zircon*, *elixir*, *natron*, and others— 25) … the importance of this period for the history of 26) … One of the leading ideas of 27) … Arabic alchemy was the theory that all metals were formed of 28) … and 29) … in various proportions and that 30) … those proportions could transform the metal under study—even to produce silver or gold from lead or iron. Not every alchemist, however, believed in the possibility of such transmutations.

Later, scholars in Christian western Europe learned of ancient Greek and early medieval Arabic philosophy by translating these books into Latin. Thus, the alchemical tradition, along with the rest of the Greco-Arabic philosophical and scientific 31) …, passed to the West in the course of the 12th century. Well-known Scholastic philosophers of the 13th century, such as Roger Bacon in England and Albertus Magnus in Germany and France, wrote on alchemy.

science physician championed remedies pharmacy controversies approach investigation practitioners quantitative distinguish conferred banishing pseudo

The Renaissance saw even stronger interest in the 32) … The German-Swiss 33) … Paracelsus practiced alchemy, Kabbala, astrology, and magic, and in the first half of the 16th century he 34) … the role of mineral rather than herbal 35) … His emphasis on chemicals 36) … in and medicine was influential on later figures, and lively 37) … over the Paracelsian 38) … raged around the turn of the 17th century. Gradually the Hermetic influence declined in Europe, however, as certain celebrated feats of putative aurifaction were revealed as frauds.

It would be a mistake to think that open-minded empirical 39) … that is well integrated with theory (which is how one might define science) was absent from the history of alchemy. Alchemy had many quite scientific 40) … through the centuries, notably including Britain's Robert Boyle and Isaac Newton—heroes of the scientific revolution of the 17th century—who applied systematic and 41) … method to their (mostly secret) alchemical studies. Indeed, as late as the end of the 17th century there was little to 42) … alchemy from chemistry, either sub-

stantively or semantically, since both words were applied to the same set of ideas. It was only in the early 18th century that chemists 43) … different definitions on the two words, 44) … alchemy to the ashbin of discredited occult 45) … sciences.

2. Find the Russian equivalents to the English words and phrases

Banish alchemy to the ashbin, discredit pseudo sciences, influential emphasis on chemicals in pharmacy, to have a detailed knowledge of smth, investigation, lively controversies, to occur, chemical identity, to indicate the presence of elemental substances, to impart yellow colour to a flame, a hazardous pollutant, to develop increasingly accurate and sensitive technique, to interact with the absorbent, to separate different components, to rage around, to apply systematic and quantitative method

3. Find the English equivalents to the Russian words and phrases

Эмпирическое совершенство ювелиров, ремесленник, обрабатывать драгоценные и полудрагоценные камни, предлагать, трансформируемость, состоять из, корни, происходить, приписывать, свойства материи, отвергать, направить усилия, лечить болезни, достигать духовную чистоту, окончательное освобождение, развивать идеи, средневековые ученые, процветать, ранние свидетельства, раскрывать, мошенничество, отличать алхимию от химии, предоставлять определения.

Text 3. The scope of chemistry

1. Read about the areas of chemistry, supplementary texts can be used to get more detailed information

The days are long past when one person could hope to have a detailed knowledge of all areas of chemistry. Those pursuing their interests into specific areas of chemistry communicate with others who share the same interests. Over time a group of chemists with specialized research interests become the founding members of an area of specialization. The areas of specialization that emerged early in the history of chemistry, such as *organic, inorganic, physical, analytical, and industrial chemistry, along with biochemistry*, remain of greatest general interest. There has been, however, much growth in the areas of *polymer, environmental, and medicinal chemistry* during the 20th century. Moreover, new specialities continue to appear, as, for example, *pesticide, forensic, and computer chemistry.*

2. Read and decide what chemistry the talk is about:

organic physical biochemistry polymer inorganic

1. Modern chemistry, which dates more or less from the acceptance of the law of conservation of mass in the late 18th century, focused initially on those substances that were not associated with living organisms. Study of such substances, which normally have little or no carbon, constitutes the discipline of … chemistry.

2. **.....** compounds are based on the chemistry of carbon. Carbon is unique in the variety and extent of structures that can result from the three-dimensional connections of its atoms. The process of photosynthesis converts carbon dioxide and water to oxygen and compounds known as carbohydrates. **.....** chemistry is the largest area of specialization among the various fields of chemistry.

3. **.....** interprets the physiological processes of living organisms in terms of molecular structure and reactivity. An organism is investigated on the premise that its physiological processes are the consequence of many thousands of chemical reactions occurring in a highly integrated manner.

4. Polyethylene, nylon, and acrylics are examples of synthetic polymers. The study of such materials lies within the domain of **.....** chemistry, a specialty that has flourished in the 20th century. The investigation of natural polymers overlaps considerably with biochemistry, but the synthesis of new polymers, the investigation of polymerization processes, and the characterization of the structure and properties of polymeric materials all pose unique problems for chemists.

5. Many chemical disciplines, such as those already discussed, focus on certain classes of materials that share common structural and chemical features. Other specialties may be centered not on a class of substances but rather on their interactions and transformations. The oldest of these fields is **.....** chemistry, which seeks to measure, correlate, and explain the quantitative aspects of chemical processes. The discovery and investigation of chemical regularities, often called laws of nature, lie within the realm of **.....** chemistry.

Text 4. Chemistry and society

1. Match the words with their definitions:

to be unlimited	make (someone) anxious or worried
public mind	the action or process of getting rid of something
by-products	someone's ability to do a particular thing/ the total amount that can be contained or produced

disposal	Something that makes it possible for people to think, feel emotions, and understand things
capacity	something that is produced as a result of making something else, or something unexpected that happens as a result of something else
concern	having the greatest possible amount, number, or level
to enrich	to be emphasized or made more noticeable
underestimate	to start to be seen or to be present
result in	to start to be seen or to be present
to come to the fore	to improve the quality of something by adding something else
impact	the way you deal with or behave towards someone or something/ the use of drugs, exercises, etc. to cure a person of an illness or injury
to appear	to fail to guess or understand the real cost, size, difficulty, etc. of something
long-term	in the end, especially after a long time or a lot of effort, problems, etc.
conversion	the process of changing something from one thing to another
legitimate	allowed by law
supervise	to watch a person or activity to make certain that everything is done correctly, safely, etc.
eventually	continuing a long time into the future
treatment	a powerful effect that something, especially something new, has on a situation or person
hazardous	the act of keeping something or a supply of something until it is needed, or a supply that you keep
prevailing	often
compounds	a way in which something can be used for a particular purpose
reserves	the act of trying to achieve a plan, activity, or situation, usually over a long period of time
to anticipate	to discover the exact size or amount of something
outweigh	to change something, usually slightly, or to cause the characteristics of something to change
application	a chemical that combines two or more elements

alter	to damage something so badly that it cannot be used
measure	to imagine or expect that something will happen
frequently	existing and accepted
pursuit	dangerous (of substances, materials, activities, or conditions)
destroy	to be greater or more important than something else

2. Read the text and fill in the gaps with the words from the box.

For the first two-thirds of the 20th century, chemistry was seen by many as the science of the future. The potential of chemical products for 1) **...** society appeared to be unlimited. Increasingly, however, and especially in the **2) ...**, the negative aspects of chemistry have **3) ... .** Disposal of chemical **4) ...** at waste-disposal sites of limited capacity has **5) ...** environmental and health problems of enormous concern. The legitimate use of drugs for the medically supervised **6) ....** of diseases has been tainted by the growing misuse of mood-altering drugs. The very word chemicals has come to be used all too 7**) ...** in a pejorative sense. There is, as a result, a danger that the **8) ...** of chemical knowledge may be seen as bearing risks that outweigh the benefits.

It is easy to **9) ...** the central role of chemistry in modern society, but chemical products are essential if the world's population is to be clothed, housed, and fed. The world's **10) ...** of fossil fuels (*e.g.,* oil, natural gas, and coal) will **11) ...** be exhausted, some as soon as the 21st century, and new chemical processes and materials will provide a crucial alternative energy source. The **12) ...** of solar energy to more concentrated, useful forms, for example, will rely heavily on discoveries in chemistry. **13) ...**, environmentally acceptable solutions to pollution problems are not attainable without chemical knowledge. There is much truth in the aphorism that «chemical problems require chemical solutions». Chemical inquiry will lead to a better understanding of the behavior of both natural and synthetic materials and to the discovery of new substances that will help future generations better supply their needs and deal with their problems.

Progress in chemistry can no longer be **14) ...** only in terms of economics and utility. The discovery and manufacture of new chemical goods must continue to be economically feasible but must be environmentally acceptable as well. The **15) ...** of new substances on the environment can now be assessed before large-scale production begins, and environmental compatibility has become a valued property of new materials. For example, **16) ...** consisting of carbon fully bonded to chlorine and fluorine, called chlorofluorocarbons (or Freons), were believed to be

ideal for their intended use when they were first discovered. They are nontoxic, nonflammable gases and volatile liquids that are very stable. These properties led to their widespread use as solvents, refrigerants, and propellants in aerosol containers. Time has shown, however, that these compounds decompose in the upper regions of the atmosphere and that the decomposition products act to **17) ...** stratospheric ozone. Limits have now been placed on the use of chlorofluorocarbons, but it is impossible to recover the amounts already dispersed into the atmosphere.

The chlorofluorocarbon problem illustrates how difficult it is **18) ....** the overall impact that new materials can have on the environment. Chemists are working to develop methods of assessment, and **19) ...** chemical theory provides the working tools. Once a substance has been identified as **20) ...** to the existing ecological balance, it is the responsibility of chemists to locate that substance and neutralize it, limiting the damage it can do or removing it from the environment entirely. The last years of the 20th century will see many new, exciting discoveries in the processes and products of chemistry. Inevitably, the harmful effects of some substances will **21) ...** their benefits, and their use will have to be limited. Yet, the positive impact of chemistry on society as a whole seems beyond doubt.

3. Find the English equivalents and make sentences with them

В уничижительном смысле, нести риски, перевешивать преимущества, необходимый химический продукт, исчерпанные резервы, важнейший альтернативный источник энергии, полагаться на научные открытия, экологически приемлемые решения, с точки зрения экономики и полезности, экономически целесообразный, ценное свойство новых материалов, быть связанным с хлором и фтором, широкое применение растворителей, хладагенты и пропелленты в аэрозольных баллонах, разлагаться, восстанавливать, методы оценки, рассеиваться.

4. Answer the questions to the text

1. What was the attitude to chemistry in the 20th century and why?
2. What concerned people's mind?
3. What gave a false color to the legitimate use of drugs?
4. What danger is the talk about in the first paragraph?
5. What might happen in the 21st century?
6. How do you understand the phrase «chemical problems require chemical solutions.»?
7. What is important about the progress in chemistry?
8. What has become a valued property of new materials?

9. What does the chlorofluorocarbon problem illustrate?

10. What is the chemists' responsibility?

5. Find the information about famous chemists and answer the questions

(Marie Curie, Louis Pasteur, Michael Faraday, Alfred Nobel)

1. What do chemists do?
2. How many famous chemists do you know?
3. What are the chemists famous for?
4. What is their origin?

UNIT 5. BIOLOGY

Lead-in:

1. Why are all people interested in biology?
2. What do you know about the Red Book? What animals and plants mentioned in this book do you know?
3. How does knowledge of biology help us to keep healthy?
4. What interesting facts about plants do you know?
5. What helps us to suggest that plants can hear?

Biology

Biology is a natural science discipline that studies living things. It is a very large and broad field due to the wide variety of life found on Earth, so individual biologist normally focuses on specific fields.

The word «biology» comes from two Greek words: bio – «life» and logos – «discourse» or «study».

Biology tells us about our body: how it is constructed and how it functions. It gives us important information about other living things and how their lives affect mankind. Knowledge of biology helps you to keep healthy. It is a guide in solving many of everyday living and scientific problems.

People who are engaged in biology are called biologists. They study the secrets of living things: how they feed, breed and survive. A biologist's laboratory is a fascinating place. Biologists make great efforts to find out everything that is possible about living things by carrying out experiments. Biologists' discoveries are of great value to all mankind.

Biologists have solved many mysteries of the body. They have discovered how blood circulates, how food is digested and many other secrets of life. They are now working in different fields of biology and their studies may lead to a solution of many problems.

Biologists have made a great contribution to science. They have increased our food supply; they have developed new and better varieties of plants and animals. Scientific methods of farming have given us much more food. Biologists control many diseases. They have saved millions of lives by discovering the causes of these diseases and methods of prevention and cure. Vaccines, penicillin and sulfa are products of the biological laboratory.

The special study of plants, Botany, and animals, Zoology, are the two great subdivisions of the science of biology. Plants and animals are called organisms, and each different type of animals, and each different

type of animal or plant is called a species, so biology may also be defined as the science of organisms.

There are millions of different plants and animals, some of which are invisible to the naked eye. They exist all over the Earth's surface, often in spite of very difficult conditions – from the icy Arctic and Antarctic to the dry, baking deserts near the Equator. They also live in the seas and oceans, from the shallow water of warm tropical seas to the gloomy ocean depth.

Most people think that plants are not alive in the same sense that animals are, or that there is some fundamental difference between plant and animal life. But there is not so. Plants and animals have much in common. Their more important points of resemblance are: 1) The living substance of plants and animals is organized into protoplasm. Protoplasm is the basic material of all living systems and its general properties are fundamentally the same in each system both in plants and animals 2) Both plants and animals consist of microscopic structural units called cells 3) Certain vital processes take place in plant bodies in the same manner as in animal bodies. These processes are respiration, sensitivity, digestion, growth and reproduction 4) Both animals and plants cannot live without certain necessary conditions. All living things need water, oxygen, and food, light and proper temperature. Both plants and animals are of different shapes, sizes and colors. In fact, the differences are not as many as the likenesses. Although they are more apparent. Plants get all the energy they need to live from sunlight by the process of photosynthesis. Animals get the energy by eating plants, animals or other organisms. Most animals can move about, and they have senses, such as sight, hearing, touch, smell and taste, which plants lack. But in a recent scientists proved that plants can hear. It comes from the phenomenon of «buzz pollination» in which a bee buzzing at a particular frequency has been shown to stimulate pollen release. Other experiments have found that sounds can lead to hormonal changes in plants, influence their oxygen uptake and change their growth rates.

Biology is divided into two ways, depending upon whether the emphasis is placed: 1) on the type of organisms 2) on processes, structure and functions. With the first system there are two principal divisions: *botany*, which deals with plants and zoology, which deals with animals.

Botany is subdivided as follows:

- Bacteriology – the study of bacteria
- Mycology – the study of fungi
- Algology – the study of algae
- Bryology – the study of mosses

• Pteridology – the study of ferns and related plants

Zoology is similarly divided as follows:

• Protozoology – the study of single-celled animals
• Entomology – the study of insects
• Ichthyology – the study of fishes
• Herpetology – the study of amphibians and reptiles
• Ornithology – the study of birds
• Mummalogy – the study of mammals
• Anthropology – the study of man

1. Answer the following questions.

1. What is biology? Define it.
2. What does the word «biology» mean?
3. How are people engaged in biology called?
4. What do biologists study?
5. Why is biology of great value to mankind?
6. What are the two great subdivisions of biology?
7. Where do living things exist?
8. Are plants and animals similar in their fundamental composition?
9. What are the differences and likenesses of plants and animals?
10. How do plants and animals depend on one another?

2. Give the English equivalents and make 10 sentences with these word combinations.

1. быть здоровым
2. решать проблему
3. заниматься чем-либо
4. прилагать усилие
5. проводить эксперимент
6. иметь большую ценность
7. делать вклад
8. несмотря на
9. иметь много общего
10. как…., так..

3. Translate the text using a dictionary.

Oceans and Corals

Oceans cover more than 70 percent of the planet and contain at least as many species as live on land. Marine biologists divide ocean species into three groups: plankton, which are floating; weak swimmers; nekton, which are stronger swimmers and benthos or bottom feeders. Plankton includes both photosynthesizes called phytoplankton and

microscopic animals called zooplankton. Phytoplankton makes up the base of the marine food chain.

Limiting factors for life in the oceans include sunlight, dissolved oxygen and availability of nutrients. Because of this most sea creatures live within 30 feet (9 m) of the water's surface. When water is cloudy, sunlight cannot penetrate very far, which means less photosynthesis can take place.

Humans have had a large impact on ocean life. For instance, modern fishing techniques are extremely destructive. Populations of many species of fish have declined precipitously because of overfishing. Some fishing techniques destroy habitat on the bottom of the ocean where coral, seaweeds and bottom feeders live.

Marine ecosystems along coasts, where seawater comes together with fresh water from rivers and other runoff, are often very productive and full of life. One such ecosystem, the sea grass meadow, is an important part of many coastal water zones.

Sea grass serves as a nursery for minnows and mollusks, which in turn provide food for others father up the food chain. Sea grass meadows, with their strong roots, help to protect coastlines and prevent erosion. They serve to store carbon and they make coastal marine environments healthier by filtering and cycling nutrients. Sea grass can be damaged by runoff from sewers and fertilizer as well as by boat propellers.

Some parts of ocean have very high biological diversity concentrated in a small area. These places include coral reefs and kelp forests, sometimes called rain forests of the ocean because they concentrate biodiversity and are also fragile and vulnerable to destruction.

Coral reefs are made by colonies of small soft creatures called polyps, which secrete a calcium carbonate (limestone) house. The brilliant colours of the most shallow-water polyps generally come from microscopic algae that live in their tissues. A coral reef has so many niches and crevices that it is a favorite habitat for a great number of fish, anemones, starfish and crustaceans.

In addition to supporting biodiversity and being a nursery for many ocean fish and other creatures, coral reefs protect about 15 percent of the world's coastlines from erosion. This is particularly important for many island nations that would not exist without their protective barriers. Coral reefs are very fragile because they grow slowly and can be sensitive to changing conditions. The familiar shallow reefs are threatened by warming seas, as well as destructive fishing and tourism practices. Deep-water reefs, which appear to be as brilliant and diverse as

their shallow-water cousins yet are only just being explored are often destroyed by fishing trawlers.

4. Your friend states botany is not biology. Explain his mistake to him. Discuss the ways in which living things differ from lifeless objects.

5. Read the text and talk about the main points of this text.

The Beginning of Life

Almost three-quarters of your body is water. In your blood scientists have found all the chemicals and minerals that make up sea water. Why? Many scientists think that all life began in the waters of the sea.

For two billions years the Earth had no living thing on it. Then, sunlight acted upon certain chemicals in the warm mud on the seafloor near the land. Lifeless chemicals joined together. They formed a new kind of molecule or particle. These molecules had new powers: they could grow by taking food. Life had begun.

The first bits of living matter were tiny pieces protoplasm. Protoplasm was in the form of tiny shapeless spots. After a long time, those tiny spots of protoplasm began to change. One type began to make food by using the energy of the sun. This type developed into the first plants, others lived by eating plants or other organisms. These were the first animals.

These shapeless bits of living matter or as we call it protoplasm, changed their form very many times! At least, may be 50 million years ago, after real life had begun, a new form of protoplasm appeared. In the new form only one part of the protoplasm did the work of getting food. Another part, the nucleus, divided the protoplasm into two. In this way a new being was formed. This was a cell. With the first cell modern life appeared because all plants and animals are made up of cells – from single-celled animals to man.

The sea gave birth to living matter. It also played a part in bringing many of the great changes, written in stones. Scientists know the earliest forms of life because they study the simple forms of sea life today.

The sea had risen and fallen for billions of years before the first one-celled beings appeared. Scientists call those first single-celled beings protozoa. After this great event another long period went by. Very many soft-bodied animals appeared in the sea; they tried to eat each other. Little by little, in the terrible struggle for life some of the soft-bodied animals developed hard shells.

For 75 million years, the great amount of hard-shelled animals lived in warm seas. Then came one of the greatest changes in the level of the

ocean. Waters fell back from the land, leaving great dry beds of seafloor, which became rock. Scientists find shells of dead sea-creatures between rock layers even today.

Later a new kind of sea creatures appeared. It was the fish. It could swim and it had a bony framework inside the body.

Again the sea changed its level, leaving lakes and marshes behind. Life in the marshes had to change or die out. Sea plants changed and some began to live in the open air.

A few fish learned to get oxygen from the air, they grew feet, not fins. They could live on dry land and returned to the sea only to lay eggs and to raise their young. They were the first amphibians. The amphibians continued to live on the edge of water. But when they grew up they stayed on the land.

Then the sea began again very slowly to cover the edge of the land. Some amphibians returned to the sea. Others learned to lay eggs and raise their young on dry land. They were a new form of life. They were all reptiles from tiny lizards to dinosaurs.

Strange animals of a new kind appeared in the marshes. They carried their young in their bodies and fed them with the milk from their own bodies. Their bodies stayed at about the same temperature in cold or warm weather. They were the first mammals.

From these earliest beginnings all the living plants and animals on land and sea have come: birds, insects, fish and mammals. They all have been developed from the first tiny bits of matter that could eat, grow and reproduce.

REVISION

1. Answer the following questions.

1. Why do scientists think life began in the sea?
2. How did life begin?
3. What powers had the first molecules?
4. What were the first bits of living matter?
5. What way was a cell formed?
6. Why did the modern life appear with the first cell?
7. Why do scientists know the earliest forms of life?
8. What was the difference of fish from other sea creatures?
9. How were amphibians formed?
10. How did reptiles appear?

2. Give the English equivalents:

1. заботиться о
2. обеспечивать кого-либо чем-либо

3. питаться
4. вызывать болезнь
5. зависеть от
6. прослеживать
7. классифицировать по
8. делить на
9. сомневаться в
10. состоять из

3. Form nouns from the following verbs and translate them into Russian.

to construct, to engage, to include, to develop, to specialize, to resemble, to digest, to provide, to discover, to contribute, to divide, to investigate, to know, to depend, to vary, to acquaint, to define, to sense, to estimate, to differ, to solve, to describe, to grow, to classify, to reproduce.

4. Translate the following text into English.

1. Биология – наука о живых организмах. Она изучает тайны живой природы: как устроены живые организмы, как они функционируют. Результаты исследований биологов имеют большое значение для развития многих отраслей науки.

2. Исследования биологов помогают решить многие проблемы современной науки. Они помогают понять взаимосвязь между всеми организмами и окружающей средой. Определение сущности жизни – одна из основных задач общей биологии.

3. Биология изучает жизненные процессы как у животных, так и у растений. Эти два больших подразделения в биологии называются ботаникой и зоологией. Как животные, так и растения не могут жить без воздуха, воды, пищи и света. Одинаковые жизненные процессы имеют место как у животных, так и у растений. Эти процессы называются дыханием, пищеварением, ростом и размножением.

4. Очень важный принцип живых организмов – это способность реагировать на внешние раздражения. Животные реагируют на внешние раздражения через нервную систему и органы чувств. Растения также приспосабливаются к окружающей среде и реагируют на внешние воздействия. Однако механизм ответной реакции у растений сильно отличается от животных и все еще недостаточно изучен. При всем объеме имеющейся информации описание некоторых видов растений и животных до сих пор неполное. Ученым предстоит раскрыть еще много тайн, и они сходятся во мнении о важности применения системного подхода в биологии при изучении процессов в живых организмах.

5. Discuss why it is important to study living things with great care and how men use plants and animals.

GLOSSARY

Unit 1

breakthrough	прорыв, серьезное научное открытие
carry out / conduct / do an experiment	проводить эксперимент
chart	график
collect data	собирать данные
contribution	вклад
create prove disprove undermine / challenge a theory sustain support weaken	создать доказать опровергнуть ставить под сомнение теорию подтвердить фактами или данными подтвердить ослабить
determine	определить
discovery	открытие, находка
do research	проводить исследование
empirical	эмпирический (основанный на опыте)
evidence	доказательство
field of science	научная область
findings	результаты
hypothesize	предполагать, выдвигать гипотезу
invention	изобретение
make observations	проводить наблюдения
make predictions	прогнозировать
observable phenomena	наблюдаемые явления
peer review	экспертное рецензирование, экспертная оценка коллег
practical use	практическое применение
publish results in a scientific journal	опубликовать результаты в научном журнале
report	отчет, доклад

reproducibility of results	воспроизводимость результатов
reveal	обнаруживать
scientific knowledge	научные знания
scientific method	научный метод
scientific records	научные записи, документы, архивы
scientific works	научные работы, труды
technological advance	научно-техническое достижение, технологический прогресс
valid results	достоверные результаты

Unit 2

acceleration	ускорение
accumulation of knowledge	накопление знаний
arise	появляться, происходить
buoyancy	плавучесть
calculus	исчисление
cause	причина, вызывать
challenge	вызов, испытание, сложная проблема
collide	сталкиваться
complete system of mechanics	завершенная система механики
conceive	придумать
constitute	составлять
continuous wave	непрерывная волна
contribution	вклад
derive from	получать
devised	разрабатывать, изобретать
displacement	перемещение, сдвиг
distinguish	различать
eclipse	затмение
equation	уравнение
equilibrium	равновесие
experimentation	экспериментирование

explode	взрываться
force	сила
foreshadow	предвещать, предзнаменовать
gravity	сила тяжести, тяготение
heliocentric system	гелиоцентрическая система
heterojunction	гетеропереход
induction	индукция
infinite	бесконечный
insight	понимание, интуиция
intersect	пересекаться
investigation	исследование
kick-start	зд. положить начало
lay the foundation	заложить фундамент
lead to	приводить к
masterpiece	шедевр
minute	зд. мельчайший
nuclear	ядерный, атомный
nucleus	ядро
occur	происходить
originate	происходить
particle	частица
phenomenon (pl. phenomena)	явление
quantum (pl. quanta)	квант
quantum-relativistic structure	квантово-релятивистская структура
quark	кварк
radiate	излучать
revealed	открывать, обнаруживать
scholar	ученый
semiconductor	полупроводник
single crystal	монокристалл
singularity	сингулярность
special theory of relativity	специальная теория относительности

squash, squeeze	сжимать
subatomic particle	субатомная частица
superconductivity	сверхпроводимость
supernova (pl. supernovae)	сверхновая
thermodynamics	термодинамика
transistor	транзистор
trigger	зд. вызывать
uranium fission	расщепление урана
usher	возвестить, объявить
velocity	скорость

Unit 3

addition	сложение
angle	угол
applied and pure mathematics	прикладная и теоретическая математика
congruence	конгруэнтность, равенство (термин, употребляемый для обозначения равенства отрезков, углов, треугольников и др. фигур и тел в элементарной геометрии)
to be congruent	быть конгруэнтными, равными
correlation between	зависимость , взаимосвязь между
curve	кривая
to calculate	вычислять, подсчитывать
division	деление
digit ;number	цифра ;число
to do sums	решать примеры
to deduce	выводить (формулу, теорему ,заключение)
to determine the value of the variable	определить значение переменной
to extract root	извлекать корень
equation	уравнение
exponent	показатель степени

integer	целое число
integer-valued function	целочисленная функция
to lead to	привести к
multiplication	умножение
to measure distance and length	измерять расстояние и длину
polygon	многоугольник
prime number	простое число
proof	доказательство
property	свойство, показатель
quantity	количество, величина
real number	действительное число
random event	случайное событие
rectangle	прямоугольник
to raise to a power	возводить в степень
space	пространство
spatial	пространственный
to simplify	упрощать
to solve problems	решать задачи
to solve algebraic expressions	решать алгебраические выражения
subtraction	вычитание
sets	множества
side	сторона
tangent	касательная
triangle	треугольник
unknown quantity	неизвестная величина
volume	объем

Unit 4

alchemy	алхимия
alcohol	спирт
alkali	щёлочь
alloys of gold and silver	сплав с золотом и серебром
amalgamating	амальгамировальный

amalgamate	соединять со ртутью
assayer	химик-лаборант
assaying	количественный анализ
atomist	последователь теории атомизма
atom	атом; мельчайшая частица
banish	избавляться
brass	желтая медь; латунь
bronze	бронза
coal	каменный уголь; уголь
composition	соединение; состав (химический)
controversy	разногласие; противоречие
copper	медь
destruction	разрушение
distilling	дистиллирующий
DNA (deoxyribonucleic acid)	ДНК дезоксирибонуклеиновая кислота
dyes	красители
elixir	эликсир
emerge	появляться; возникать
fraud	обман; умышленное искажение либо неверное толкование фактов
gilding	золочение
gold	золото
herbal remedies	растительные лечебные средства
hydrochloric acid	хлористоводородная кислота
impart	придавать; наделять
impose	вынуждать; навязывать
integrate	интегрировать; объединять
intermingle	смешивать; перемешивать
irreducible	невосстанавливающийся

lacquering	лакирующий
lead	тетраэтилсвинец
magnetite	магнетит (Fe3O4); магнитная окись железа (Fe3O4)
matter	вещество; материя
metalsmiths	мастер по художественной ковке металла
moist	смачивать
natron	углекислый натрий
nitric acid	азотная кислота; азотистая кислота
oxygen	кислород
particle	частица
permanence	неизменность; стабильность
preclude	предотвращать; устранять
putative	мнимый; предполагаемый
robust	прочный
scholar	исследователь
smelting	плавильный
steels	металлы
sublimate	сублимировать; очищаться
sulfuric acid	серная кислота
tin	олово
vinyl	винил
zircon	цирконий

Unit 5

acquaintance	знакомство
acquire	приобретать
algae	водоросли
attach	прикреплять
blood circulation	кровообращение
bottom	донный; обитающий на дне
catch	улов; добыча

contribute (to)	способствовать; содействовать
creature	существо
cryptogam plant	споровое растение
digest	переваривать(ся) (о пище)
fern	папортник
fertilizer	удобрение
horsetail	хвощ
investigation	исследование
kelp forests	ламинарии
level	выравнивать
moss	мох
niches and crevices	ниши и щели
nursery	питомник; инкубатор
phanerogamic plants	фанерогамный; растение с семенами
poisonous	ядовитый
pollination	опыление
reproduction	размножение
resemblance	сходство; подобие
secrete	выделять
sewers	канализация
species	вид
shallow bottom	мелководье; отмель
suctorial	сосущий
tissues	ткани
vulnerable	уязвимый

REFERENCES

1. Britannica Dictionary. URL: https://www.britannica.com/dictionary (дата обращения: 26.02.2023).

2. Cambridge Dictionary. URL: https://dictionary.cambridge.org/ (дата обращения: 26.02.2023).

3. *Ekinci Deniz* Chemical Biology. – Turkey: InTech. 2012. – 444 p.

4. *Gerczia Timea, Pattison Scott* Biochemistry Laboratory Manual for Undergraduates. – Berlin: Walter De Gruyter Inc., 2014. – 174 p.

5. *Kozharskaya E., Mc Nicholas, K. Macmillan* Guide to Science. Student's Book. – Oxford: Macmillan ELT, 2008. – 127 p.

6. Oxford Learners Dictionary. URL: https://www.oxfordlearners-dictionaries.com (дата обращения: 26.02.2023).

7. Seven principles for scientific publishing. URL: https://council.science/current/news/seven-principles-for-scientific-publishing/ (дата обращения: 26.02.2023).

8. Space Place. What is science? URL: https://spaceplace.nasa.gov/science/en/ (дата обращения: 26.02.2023).

9. The 10 Most "OMG" Science Discoveries of 2022. URL: https://bestlifeonline.com/news-most-omg-science-discoveries-2022/ (дата обращения: 26.02.2023).

10. The early Universe. URL: https://home.cern/science/physics/early-universe (дата обращения: 26.02.2023).

11. The famous people. Chemists. URL: https://www.thefamouspeople.com/chemists.php (дата обращения: 26.02.2023).

12. *Thornton S., Rex A.* Modern physics for scientists and engineers. – Boston: Brooks/Cole, Cengage Learning, 2013. – 615 p.

13. Tomorrow's technology depends on today's physics. URL: https://singularityhub.com/2015/01/28/tomorrows-technology-depends-on-todays-physics/ (дата обращения: 26.02.2023).

14. *Urone P., Hinrichs R.* Physics for high school students. – Houston: OpenStax, Texas Education Agency. 2020. – 850 p.

15. What is science? URL: https://www.science.org.au/curious/people-medicine/what-science (дата обращения: 26.02.2023).

16. *Еремеева Г. Р.* English for Students of Mathematics: учебное пособие / Г. Р. Еремеева, Н. А. Сигачева, Ф. Б. Ситдикова. – Казань: КФУ, 2022. – 184 с.

17. *Кирпичникова А. А., Сосновская Г. И.* Professional English for Biology and Medicine Students: учебно-методическое пособие. – Казань: Изд-во Казан. Ун-та, 2017. – 110 с.

18. *Назарова Н. А., Панасенко Е. В., Толстых О. М.* Professional English in Use: Mathematics and Physics. – Омск: ОмГПУ, 2018. – 124 с.

19. *Полубиченко Л. В.* Английский язык для направления «Биология»: учебное пособие. – Спб.: Academia, 2020. – 256 с.

20. *Пушкина Е. Н.* English for Mathematicians and Information Technologies Learners. Английский для студентов, изучающих математику и информационные технологии: учебно-методическое пособие. – Нижний Новгород, ННГУ, 2019. – 89 с.

21. *Сафроненко О. И., Грушко Е. И., Петросян К. С.* Learn the English of physics. – Ростов-на-Дону: кмц ЮФУ, 2014. – 157 с.

22. *Чвягина Т. В., Невская Е. А.* Биология на английском: методические указания. – Ярославль: ЯрГУ, 2019. – 43 с.

23. *Шевырдяева Л. Н., Кожарская Е. Э.* English for Life Sciences: Plants and Animals: рабочая тетрадь. – М.: Издательство КДУ, 2022. – 140 с.

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